

Improving Answer Groundness in Large Language Models through LoRA Fine-Tuning and Retrieval-Augmented Generation

*A Comparative Evaluation on GovReport-QS and QASPER across Four
Dimensions of Groundness*

Master's Thesis

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Abstract

Large language models (LLMs) have achieved remarkable performance in question answering (QA) but remain susceptible to hallucinations, posing a significant challenge in knowledge-intensive applications. This thesis investigates how two complementary augmentation strategies, retrieval-augmented generation (RAG) and parameter-efficient fine-tuning (LoRA), individually and jointly affect groundness, a multidimensional construct encompassing context relevance, answer relevance, faithfulness, and attribution.

Using a comparative experimental design, four model configurations were evaluated: a base model, a RAG-only model, a LoRA-only model, and a combined RAG+LoRA model. These models were assessed on two long-document QA datasets, GovReport-QS and QASPER, using a multi-level evaluation framework that integrated automated metrics with statistical analysis at both the sample and document levels.

The findings reveal that RAG consistently improves context relevance and faithfulness relative to the base model. LoRA's effects are domain-dependent, enhancing groundness on the structured GovReport-QS dataset but reducing faithfulness on the heterogeneous QASPER dataset. The combined RAG+LoRA approach yields synergistic benefits on GovReport-QS, while on QASPER it presents a coverage-consistency trade-off, improving recall and attribution at the expense of faithfulness.

This study makes several contributions: it empirically validates the multidimensional nature of groundness, provides a rigorous evaluation framework for assessing LLM reliability, and offers practical, domain-specific guidance for deploying grounded QA systems. The results highlight both the promise and the complexities of combining augmentation strategies to enhance LLM trustworthiness.

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1.Introduction

Large language models (LLMs) have rapidly advanced the state of natural language processing and are increasingly deployed in knowledge-intensive tasks such as question answering (QA). Despite their impressive fluency, a persistent limitation is the tendency to produce hallucinations—outputs that are fluent and confident but factually unsupported. This issue is particularly critical in domains where reliability and verifiability are essential, such as policy, law, and scientific research. In response to these challenges, the concept of groundness (sometimes also referred to as groundedness) has gained prominence as the positive counterpart of hallucination, emphasizing that model answers should be relevant to the input question, faithful to retrieved or provided evidence, and transparent in attribution. Understanding how different augmentation strategies, such as retrieval-augmented generation (RAG) and parameter-efficient fine-tuning, affect groundness is therefore essential for both advancing QA methodology and ensuring safe deployment of LLM-based systems.

1.1 Problem Formulation

1.1.1 Superior Scope and Topic Delimitation

This study focuses on evaluating the effectiveness of LoRA fine-tuning and Retrieval-Augmented Generation (RAG) in improving the groundness of large language model (LLM) outputs across four core dimensions: Context relevance, Answer faithfulness, Answer relevance, and Attribution/Transparency. The primary dataset used is GovReport-QS, which contains question-answer pairs grounded in structured government documents. A secondary dataset, QASPER, is used for comparative validation to test the generalizability of findings across domains.

The scope is limited to experiments conducted on the Mistral 7B model, using a controlled set of four model configurations (M0–M3) to isolate the effects of LoRA and RAG, both separately and jointly. The evaluation relies on automated metrics including RAGAS, e5-base-v2, and AIS, rather than human evaluation, in order to ensure replicability and scalability.

This research deliberately excludes dimensions such as human preference, fluency, or completeness, focusing strictly on groundness—defined as the alignment between generated content and evidence. Furthermore, the study only explores single-turn QA tasks in long-document settings and does not investigate multi-turn dialogue, intent classification, or model generalization beyond the two datasets.

1.1.2 Research Questions

How fine tuning and RAG approaches will influence the language models in improving the groundness of questions and answers?

Sub question:

1.RQ1: The Effect of RAG

How does augmenting a model with retrieval (M1) affect context relevance and answer faithfulness compared to a base model without retrieval (M0)?

2.RQ2: The Effect of Fine-tuning

How does augmenting a model with retrieval (M1) affect context relevance and answer faithfulness compared to a base model without retrieval (M0)?

3.RQ3 (Synergy).

How does combining retrieval augmentation and LoRA fine-tuning (M3) influence answer groundness—particularly faithfulness and attribution—compared to their individual contributions (M1 and M2)?

1.2 Thesis Outline

To answer the research and supportive questions stated above, the thesis is structured into multiple chapters. In Chapter 2, a review of the related literature on Retrieval-Augmented Generation (RAG), fine-tuning, and groundness evaluation in question answering systems is provided. This sets the stage for Chapter 3, where the conceptual framework is introduced. The framework establishes a general understanding of RAG and LoRA fine-tuning, explains

the technical foundation of the Mistral 7B base model, and defines the four dimensions of groundness: context relevance, answer faithfulness, answer relevance, and attribution. This is followed by Chapter 4, which first describes how the research is conducted by introducing the Research Philosophy and the experimental design. Additionally, a methodological explanation of the datasets (GovReport-QS and QASPER), the four model configurations (M0: Mistral 7B, M1: M0+RAG, M2: M0+LoRA fine-tuning, M3: M0+RAG+LoRA fine-tuning), and the evaluation steps using metrics such as RAGAS-faithfulness, RAGAS-answer_relevancy, e5-base-v2, and AIS are included as well. In Chapter 5, the results of the experiments that were outlined in the methodology are presented, addressing the effects of RAG (RQ1), Lora Fine-tuning (RQ2), and their synergy (RQ3), which is followed by a discussion of the results in Chapter 6. Besides that, the discussion also includes RQ answers, implications, limitations, future work and recommendations. To summarize the thesis, a conclusion is provided in Chapter 7.

2. Literature Review

2.1 Large Language Model and QA

Early developments in question answering (QA) systems encompassed a variety of approaches, including information retrieval (IR)-based methods, rule-based and knowledge-based systems, and template-driven techniques. As reviewed by Calijorne Soares & Parreiras (2020), these systems typically operated over restricted domains, relying on handcrafted rules, linguistic patterns, or predefined templates. While they demonstrated reasonable effectiveness in specific applications, their dependence on manual design limited adaptability and made it difficult to scale toward open-domain QA tasks.

The advent of machine reading comprehension (MRC) marked a shift toward extractive QA, where models identify answer spans directly within passages. A milestone in this development was the introduction of the Stanford Question Answering Dataset (SQuAD), which provided over 100,000 crowd-sourced questions paired with passages from Wikipedia and established extractive span prediction as a benchmark task (Rajpurkar et al., 2016). Building on this benchmark, the emergence of BERT demonstrated that large-scale pre-

training of deep bidirectional Transformers could achieve state-of-the-art results on SQuAD and similar datasets, significantly advancing extractive QA performance (Devlin et al., 2018). However, despite their effectiveness in these controlled settings, such models remained constrained by dataset-specific training and struggled to generalize to more open-domain scenarios.

The introduction of the Transformer architecture by [Vaswani et al. \(2023\)](#) fundamentally reshaped natural language processing by replacing recurrent and convolutional structures with a self-attention mechanism. This innovation enabled large-scale pre-training and parallelized sequence modelling, laying the foundation for modern large language models (LLMs). Building on this foundation, models such as BERT and the GPT series catalysed the emergence of LLMs as general-purpose reasoning systems. [Zhao et al., \(2025\)](#) review this evolution, outlining four key pillars of LLM development—pre-training, adaptation tuning, utilization, and evaluation—while also identifying persistent challenges such as high training costs, lack of transparency, and hallucination.

The emergence of LLMs has transformed QA by enabling generative approaches that go beyond span extraction. Instead of locating a fixed answer span, LLMs can synthesize responses through reasoning and contextualization, positioning them as general-purpose engines for information-seeking tasks. This paradigm shift is captured in recent surveys such as [\(Zhao et al., 2025\)](#), who propose a unified framework for understanding LLM-based QA agents. They divide the workflow into four stages: planning (decomposing complex queries into sub-tasks), information retrieval (gathering relevant evidence from external sources), tool use (interacting with search engines or databases), and answer generation (synthesizing retrieved information into a final response). In this agentic perspective, LLMs function not merely as text generators but as orchestrators capable of coordinating reasoning, retrieval, and tool interaction.

However, as [Zhao et al., \(2025\)](#) emphasize, hallucination remains a fundamental limitation, as models may generate fluent but unsupported content. This persistent challenge underscores the need for augmenting LLMs with additional context, which will be the focus of the next section.

2.2 Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) has emerged as a central paradigm for overcoming the core limitations of large language models (LLMs), particularly their reliance on static, pre-trained knowledge and their tendency to hallucinate. [Lewis et al., \(2021\)](#) first demonstrated that coupling retrieval with generation enables models to access external evidence, thereby grounding outputs more effectively in factual information. Building on this foundation, recent surveys have systematized RAG into a three-stage framework—retrieval, augmentation, and generation—clarifying how retrieved information functions as contextual input to the LLM and highlighting the challenges and variants of this paradigm ([Gao et al., 2024](#)). Within this framework, the “context” that guides an answer is not simply drawn from a model’s parameters but is dynamically constructed by retrieving and augmenting information from external sources such as vector databases or document collections. [Shuster et al., \(2021\)](#) further show that retrieval augmentation substantially reduces hallucinations in open-domain dialogue: on benchmarks such as Wizard of Wikipedia and CMU_DoG, RAG-based models generated significantly fewer hallucinated responses compared to non-retrieval baselines.

An alternative perspective extends the notion of context beyond retrieval. [Yu et al., \(2023\)](#) proposed *GenRead*, a generate-then-read pipeline where LLMs themselves act as context generators by producing synthetic passages tailored to a query. These generated contexts are then used in much the same way as retrieved evidence, but they are created internally rather than drawn from an external corpus. This contrast between retrieved context and generated context illustrates the growing breadth of approaches to grounding LLM answers.

At the same time, advances in reasoning methods complement these context-focused strategies. [Wei et al., \(2023\)](#) introduced Chain-of-Thought (CoT) prompting, which elicits step-by-step reasoning chains in LLM outputs. While CoT does not provide new knowledge, it improves how models use either retrieved or generated context by structuring their reasoning process. From this perspective, RAG and CoT address different but complementary challenges: RAG enriches the source of knowledge, whereas CoT enhances the use of knowledge.

Complementing these developments, [Rüdel & Leidner, \(2023\)](#) offer insights into deploying hybrid chatbots in real-world, business-critical environments. They describe a case study integrating a commercial rule engine with a neural chatbot using RAG to ensure that assistants only answer based on verified, company-provided data—thus maintaining high control over output content and minimizing hallucination. Their *Kauz platform* architecture enables human editors to steer the assistant through tools like prompt design, metadata annotation, and configuration of generation safeguards, effectively controlling when the LLM generates text versus relying purely on structured rule-based knowledge.

Taken together, this line of research traces a clear trajectory: from stand-alone LLMs, to retrieval-augmented systems that ground responses in external knowledge, to frameworks that expand context generation itself, to models enhanced with explicit reasoning capabilities, and finally to real-world hybrid deployments that balance generative freedom with controlled, fact-based output. Within this progression, RAG emerges as a foundational mechanism for bridging the gap between static LLM knowledge and the demands of interactive, reliable QA systems.

2.3 Lora Fine-tuning

Transfer learning, as a paradigm, mitigates the limitations of conventional machine learning—wherein the training and test data are assumed to be from the same distribution—by leveraging knowledge from a related source domain to enhance performance in a target domain ([Pan & Yang, 2010](#)). This foundational principle addresses the high cost and time required for collecting and annotating large-scale datasets for every new task. Building upon this, [Zhuang et al., \(2020\)](#) conducted a comprehensive survey, reviewing over forty representative approaches and systematically categorizing them from both data and model perspectives. Their work underscores that the practical effectiveness of transfer learning is contingent upon selecting the appropriate method for a given application scenario, a conclusion supported by their experiments on diverse datasets. This systematic understanding of transfer learning provides the theoretical underpinning for fine-tuning in large language models, specifically as a form of Parameter Transfer ([Pan & Yang, 2010](#)) where knowledge is transferred through shared model parameters from a pre-trained source model to a target task.

The introduction of the Transformer architecture by [Vaswani et al., \(2023\)](#) established a fully attention-based model that enabled large-scale pretraining on massive text corpora, laying the foundation for subsequent advances in natural language processing. Building on this, [Devlin et al., \(2018\)](#) proposed BERT, which leverages deep bidirectional Transformers trained on large unlabelled corpora using masked language modelling and next-sentence prediction objectives. Crucially, BERT demonstrated that pretrained representations could be adapted to downstream tasks through fine-tuning, where the same model architecture is initialized with pretrained parameters and then trained with task-specific data. This paradigm shift highlighted fine-tuning as a general mechanism for transferring knowledge from large-scale pretraining to a wide range of NLP applications, such as question answering and natural language inference. In parallel, [Howard & Ruder., \(2018\)](#) introduced ULMFiT, which systematically established fine-tuning as a universal method in NLP by showing how a pretrained language model could be effectively adapted to diverse text classification tasks with limited labelled data. Together, these works illustrate both the practical effectiveness and methodological foundations of fine-tuning as a transfer learning strategy in NLP.

Instruction fine-tuning, also known as supervised fine-tuning (SFT), is the process of further training large language models (LLMs) on datasets of (instruction, output) pairs to better align the model's behavior with human intent ([Zheng et al., 2023](#)). This approach is crucial for bridging the gap between an LLM's pretraining objective (next-token prediction) and its practical, user-facing goal of accurately executing tasks. In a landmark study, [Wei et al., \(2022\)](#) demonstrated that instruction-tuned models like FLAN, trained on over 60 NLP tasks, substantially outperform their untuned counterparts. FLAN's exceptional generalization to unseen tasks underscored instruction tuning's capacity to transform an LLM into a competent zero-shot learner. While [Zhang et al., \(2025\)](#) surveyed the methodology and performance factors of instruction tuning, they also highlighted areas needing deeper exploration, such as optimal dataset construction, suggesting clear avenues for future research. Additionally, the high training and storage costs associated with fine-tuning increasingly massive models present a significant challenge that necessitates the development of more parameter-efficient methods. This suggests clear avenues for future research.

Large language models (LLMs) have become so massive that full fine-tuning—updating all parameters for a specific task—is computationally and memory-prohibitively expensive. To

solve this, Hu et al., (2021) introduced LoRA (Low-Rank Adaptation), a parameter-efficient fine-tuning (PEFT) method. LoRA freezes the original pretrained weights and injects small, trainable low-rank decomposition matrices (A and B) into the Transformer layers. These matrices approximate the full weight update ($\Delta W = BA$), drastically reducing the number of trainable parameters. For example, on models like GPT-3, LoRA reduces the number of trainable parameters by up to 10,000 times and cuts GPU memory usage by about 3 times. Crucially, it achieves performance on par with or even better than full fine-tuning, while adding no extra latency during inference. The paper also provides empirical evidence that fine-tuning updates have a low "intrinsic rank," explaining why this low-rank approximation is so effective. Moreover, recent work has demonstrated the effectiveness of LoRA in downstream QA tasks: for instance, the ChatKBQA framework integrates fine-tuned LLMs with knowledge base question answering, showing that LoRA-tuned models can generate candidate answers and then retrieve supporting knowledge to improve accuracy and grounding (Luo et al., 2024)

2.4 Groundness

2.4.1 Hallucination & Groundness

Despite recent advances such as Retrieval-Augmented Generation (RAG) and parameter-efficient fine-tuning methods like LoRA, hallucination remains a persistent and fundamental challenge for large language models (LLMs). Hallucination broadly refers to cases where models generate fluent and plausible text that is nevertheless either factually incorrect or unfaithful to the provided context (Huang et al., 2025). This issue is particularly critical for question answering (QA), as it directly undermines reliability, can propagate misinformation, and poses risks in knowledge-intensive or high-stakes applications (Ji et al., 2023).

Scholars have proposed several taxonomies of hallucination. Huang et al., (2025) distinguish between factuality hallucinations and faithfulness hallucinations, while Ji et al., (2023) describe broader categories such as *intrinsic* versus *extrinsic* hallucinations across natural language generation tasks. Building on this, Maynez et al., (2020) provided a widely adopted definition in abstractive summarization, identifying hallucinations as content that is either unfaithful to or unfactual with respect to the source. They further divide hallucinations into

two types: intrinsic, where the generated text directly contradicts the source, and extrinsic, where it introduces unsupported information. Their large-scale human evaluation revealed that hallucinations were pervasive, occurring in more than 70% of single-sentence summaries, thus highlighting the gap between surface fluency and true grounding.

In response to these challenges, the concept of *groundness* (sometimes also referred to as *groundedness*) has gained prominence as the positive counterpart of hallucination. Groundness refers to a model's ability to produce outputs that are accurate, verifiable, and supported by reliable source information. It emphasizes not merely the absence of hallucinations, but the requirement that generated content can be explicitly traced back to evidence. ("Understanding Groundness in Language Model Chat Applications.," 2025) Academic studies illustrate this distinction: [Stolfo, \(2024\)](#) shows that even when LLMs produce factually correct answers, many sentences remain ungrounded if they lack direct support from retrieved documents; [Adlakha et al., \(2024\)](#) similarly argue that correctness must be complemented by faithfulness, since responses often fail to remain grounded in their input context.

Beyond academia, groundness has also become an operational priority in industry. Deepset ("RAG LLM Evaluation: Groundedness," 2024) defines groundness as the degree to which RAG outputs are supported by retrieved documents, conceptualizing it as the opposite of hallucination. Microsoft ("Groundedness in Azure AI Content Safety.," 2024) integrates groundness detection into its content safety framework, verifying whether LLM outputs align with provided sources, particularly in sensitive applications such as medical summarization and customer service. Databricks ("LLM Judge Metrics for Agent Evaluation," 2024) likewise incorporates groundness into evaluation pipelines for enterprise AI agents, where automated "LLM judges" assess outputs along dimensions including groundness and correctness.

Collectively, these perspectives demonstrate that hallucination remains a central obstacle to reliable QA, while groundness has emerged as a unifying goal across both research and industry—shifting the focus from merely avoiding errors to ensuring that answers are faithfully anchored in evidence.

2.4.2 Dimensions of Groundness

Having established groundness as a core requirement for reliable LLM outputs, prior research has decomposed it into distinct dimensions that can be systematically evaluated.

Evaluating context relevancy—whether the information retrieved for a query is pertinent to generating the final answer—is foundational for ensuring groundness in RAG systems. [Saad-Falcon et al., \(2024\)](#) introduced ARES, an automated evaluation framework that assesses context relevance alongside answer faithfulness and relevance by training lightweight language-model-based judges on synthetic data, enabling efficient, scalable assessment of RAG components with minimal human annotation. Complementarily, the RAGAS toolkit formalizes context relevancy as a dedicated metric, evaluating whether retrieved documents align with the query—operationalized through measures like context precision, context recall, and context utilization (“Faithfulness Metric in RAGAS,” 2023). These tools highlight that a RAG system’s reliability depends not merely on generation but on retrieving truly relevant context—making context relevancy a critical dimension of groundness. Together, these approaches demonstrate that grounding begins with retrieving truly relevant evidence, which forms the basis upon which other dimensions, such as faithfulness and relevance, can be meaningfully assessed.

Within the broader evaluation of groundness, faithfulness has been widely recognized as a critical dimension for assessing the reliability of large language model (LLM) outputs. Faithfulness refers to the extent to which generated responses accurately reflect and remain consistent with the supporting evidence provided. For example, [Saad-Falcon et al., \(2024\)](#) proposed ARES, an automated evaluation framework for retrieval-augmented generation (RAG) systems, which incorporates faithfulness as one of its key evaluation criteria, leveraging lightweight LLM judges trained on synthetic data and validated with limited human annotations. Similarly, the RAGAS toolkit conceptualizes faithfulness at the claim level, measuring the proportion of factual claims in a generated response that can be directly inferred from retrieved context (“Faithfulness Metric in RAGAS,” 2023). Complementing these system-oriented perspectives, [Adlakha et al., \(2024\)](#) evaluated instruction-following models for question answering along the dual axes of correctness and faithfulness, showing that traditional metrics such as exact match or F1 are insufficient for

capturing whether answers truly adhere to the source material. Together, these works establish faithfulness as a core dimension of groundness, essential for ensuring that model outputs are not only factually accurate but also verifiably aligned with their evidence base. These findings collectively position faithfulness as complementary to context relevance: while relevant documents must be retrieved, responses must also remain strictly consistent with the evidence they provide.

In the evaluation of retrieval-augmented generation (RAG) and instruction-following question-answering systems, answer relevance measures how well the generated response directly addresses the user's query in alignment with the provided context. Answer relevance thus captures a different but equally crucial aspect of groundness, moving beyond alignment with context to ensure that generated outputs directly satisfy the user's query.

Saad-Falcon et al., (2024) introduced ARES, an automated RAG evaluation framework, which includes answer relevance as a key assessment dimension alongside context relevance and faithfulness. ARES uniquely combines lightweight LLM judges trained on synthetic data with a small human-annotated validation set to accurately assess whether a response is pertinent to the query—achieving high evaluation accuracy with minimal annotation efforts. Complementing this, Adlakha et al., (2024) dissect QA model outputs along both correctness and faithfulness, implicitly underscoring answer relevance as a litmus test for whether generated responses satisfy the user's information need while remaining grounded in the source. Adding to these, Zha et al., (2023) propose AlignScore, a unified factual consistency metric based on alignment between input and output text. While primarily focused on factuality, AlignScore's alignment mechanism also reflects answer relevance by evaluating semantic correspondence between a response and its source context. Together, these methodologies position answer relevance as a distinct—and vital—dimension of groundness, crucial for ensuring generated answers are not only faithful but also meaningfully responsive. In doing so, answer relevance bridges the gap between factual grounding and practical utility in QA systems.

In the realm of natural language generation (NLG), attribution refers to whether generated content can be traced back and verified against identified information sources—a critical aspect beyond mere factuality or faithfulness. Rashkin et al., (2022) introduce the Attributable

to Identified Sources (AIS) framework for measuring attribution, defining it through whether each statement in model output pertaining to the external world is supported by provided, verifiable sources. They propose a rigorous two-stage human annotation pipeline and empirically validate AIS across diverse tasks—including conversational QA, summarization, and table-to-text generation—demonstrating its efficacy as a unified metric for evaluating whether NLG outputs genuinely derive from source material. This approach underscores attribution as a foundational dimension of groundness that exposes whether model-generated statements are evidence-based rather than hallucinated or invented. Attribution complements context relevance, faithfulness, and answer relevance by demanding explicit traceability of model outputs to identifiable sources, thereby exposing whether responses are genuinely evidence-based.

Beyond traditional groundness dimensions, recent research has introduced more nuanced evaluation facets to assess how long-form generated text aligns with factual support and robustness. The FACTSCORE metric, introduced by Min et al., (2023), decomposes generated content into atomic facts and computes the proportion of these facts that are verifiably supported by a trusted knowledge source. This granular approach addresses the limitations of binary evaluations by enabling partial credit when some—but not all—information is accurate (Min et al., 2023). These complementary frameworks suggest that groundness is not limited to static dimensions but also encompasses robustness-oriented properties that are particularly relevant for long-form QA.

Complementing this, Chen et al., (2023) test retrieval-augmented generation (RAG) systems against four foundational abilities critical for effective grounding but not captured by faithfulness or correctness alone. These include noise robustness (resilience to irrelevant or noisy retrieved data), negative rejection (the system’s ability to abstain when no reliable information is retrieved), information integration (the capacity to synthesize information from multiple retrieved documents), and counterfactual robustness (the ability to resist plausible but false retrieved facts). Their evaluation on the RGB benchmark reveals that current LLMs struggle significantly across these dimensions, indicating that improving RAG effectiveness demands deeper consideration of these operational properties (Chen et al., 2023).

Taken together, these studies converge on four primary dimensions of groundness—context relevance, faithfulness, answer relevance, and attribution—while also pointing toward emerging evaluation facets such as robustness and factual decomposition. These dimensions form the conceptual foundation for the evaluation framework adopted in this thesis.

2.4.3 Evaluation Methods for Groundness

Having identified the core dimensions of groundness, the next step is to outline the methods by which these dimensions can be systematically measured. Evaluation approaches can be broadly grouped into embedding-based similarity metrics, human evaluation, and LLM-as-judge paradigms, each offering distinct advantages and limitations.

Text embeddings, such as the E5 family, have become a cornerstone in evaluating various downstream NLP tasks. Developed through a two-stage process that includes large-scale, weakly-supervised contrastive pre-training on datasets like CCPairs, these models demonstrate exceptional versatility and efficiency (Cao & Wang, 2022). The E5 models have consistently delivered state-of-the-art performance in both zero-shot and fine-tuned settings. Notably, they were the first unsupervised models to surpass the BM25 baseline on the BEIR retrieval benchmark, and their performance on the MTEB benchmark further validated their effectiveness. This capability to capture nuanced semantic relationships allows practitioners to quantitatively assess model relevance and retrieval quality by comparing vector representations via similarity metrics. Building on this, frameworks such as RAGAS integrate embedding-based measures within systematic pipelines, formalizing metrics like context precision, context recall, and context utilization to evaluate whether retrieved documents and generated answers align with the query (“Faithfulness Metric in RAGAS,” 2023). These approaches highlight the centrality of pretrained representation spaces in automated evaluation, though they often fail to capture fine-grained factual consistency and can misjudge semantically plausible but unsupported answers (Saad-Falcon et al., 2024).

Human annotation has long been regarded as the gold standard for assessing the quality of open-ended text generation, since annotators can directly judge coherence, relevance, and informativeness. However, several studies point out that while human judgments provide reliable ground truth, they are costly, time-consuming, and difficult to scale to large model

families (Zheng et al., 2023)(Li et al., 2024). Moreover, inter-annotator agreement is often imperfect, introducing inconsistency into results. These challenges underscore the need for scalable alternatives that can approximate human-level judgments more efficiently.

To address these limitations, researchers have increasingly adopted LLMs themselves as evaluators. Zheng et al., (2023) propose the LLM-as-a-judge paradigm, leveraging high-performing models such as GPT-4 to evaluate open-ended tasks. Their benchmarks, MT-Bench and Chatbot Arena, show that LLM judges achieve over 80% agreement with human preferences, effectively reaching human-level consensus. Similarly, (Li et al., 2024) introduce the BenchBuilder pipeline, which constructs high-quality benchmarks (e.g., Arena-Hard-Auto) from crowdsourced prompts while relying on LLMs to both curate and evaluate responses. BenchBuilder achieves 98.6% correlation with human rankings and three times higher model separability than existing benchmarks. These findings suggest that LLM evaluators can provide scalable, cost-effective, and robust alternatives to purely human evaluation. However, while LLM-based judges reach human-level consensus, they also raise concerns regarding computational cost, scalability, and potential biases. ARES directly addresses these limitations by training lightweight LLM judges on synthetic data, thereby providing a more efficient and transparent alternative to heavyweight evaluators (Saad-Falcon et al., 2024).

Taken together, embedding-based metrics offer efficiency and scalability, human evaluation provides authoritative but costly gold standards, and LLM-as-judge approaches present a promising compromise by combining scalability with human-level reliability. In this thesis, these categories of evaluation methods inform the assessment strategy for measuring groundness across datasets and model configurations.

3. Conceptual Framework

3.1 Core concepts

This study is grounded in a conceptual framework that distinguishes between the interventions applied to large language models (LLMs) and the outcomes through which

their effectiveness is assessed. The interventions—Retrieval-Augmented Generation (RAG), LoRA fine-tuning, and their combination—represent different strategies for enhancing model performance, either through external knowledge augmentation or internal parameter adaptation. The outcome construct is defined as *groundness*, understood as the degree to which model outputs remain accurate, verifiable, and aligned with supporting evidence. By clarifying these core concepts, the framework provides a structured basis for analysing how different interventions affect multiple dimensions of groundness in question answering tasks.

3.1.1 Interventions

The interventions in this study encompass two complementary strategies for enhancing large language models in the context of question answering: Retrieval-Augmented Generation (RAG) and LoRA fine-tuning. These approaches are selected because they directly address two central challenges identified in the literature: the tendency of LLMs to hallucinate when relying solely on static parametric memory, and the prohibitive cost of full fine-tuning for adapting increasingly large models to domain-specific tasks. Together with their combined application, they define the independent constructs of the framework, each expected to influence specific dimensions of groundness in distinct ways.

Retrieval-Augmented Generation (RAG) is framed in this study as an external augmentation strategy that complements the parametric knowledge of large language models with dynamically retrieved passages from external sources (Lewis et al., 2021). By enriching the input prompt with semantically relevant documents, RAG grounds model outputs in up-to-date and verifiable evidence. Within the present framework, RAG is expected to primarily improve context relevance, ensuring that answers draw from information aligned with the query, while also supporting faithfulness by reducing reliance on unsupported parametric memory. In this way, RAG functions as a means of embedding evidence directly into the generation process, addressing one of the most persistent challenges of hallucination.

LoRA fine-tuning, by contrast, is conceptualized as an internal adaptation strategy for large language models. By introducing low-rank parameter matrices into pretrained Transformer layers while keeping the original weights frozen, LoRA enables efficient domain adaptation without the prohibitive cost of full fine-tuning (Hu et al., 2021). Within the context of long-

form QA, LoRA is expected to enhance faithfulness—by aligning generated responses more closely with gold-standard evidence—and answer relevance, by reducing off-topic or unsupported content. In this sense, LoRA represents a mechanism for tailoring model behavior toward greater reliability under task-specific supervision, complementing RAG’s evidence-augmentation function.

3.1.2 Outcome

Groundness, as discussed in the literature, has emerged both in research and industry as a central criterion for evaluating large language model outputs. It is frequently framed as the positive counterpart to hallucination, emphasizing not only the avoidance of unsupported content but also the requirement that answers be evidence-based, transparent, and verifiable. Industrial frameworks such as those developed by Microsoft, Deepset, and Databricks have operationalized groundness as a practical evaluation metric, while academic studies highlight its role in bridging the gap between surface fluency and factual reliability. Building on these perspectives, this study defines groundness in a task-specific way: it is the outcome construct through which the effectiveness of LoRA and RAG interventions is assessed, focusing on whether generated responses remain accurate, trustworthy, and explicitly linked to supporting evidence.

To make this construct analytically tractable, groundness is decomposed into four interrelated dimensions:

- **Context relevance:** whether the retrieved evidence is pertinent to the query and forms an appropriate basis for generating the answer.
- **Faithfulness:** the degree to which generated responses remain consistent with, and do not contradict, the provided evidence.
- **Answer relevance:** the extent to which the generated response directly addresses the user’s information need, beyond mere factual consistency.
- **Attribution:** whether the generated content can be explicitly traced back to identified evidence sources, ensuring transparency and verifiability.

Although attribution is not directly optimized by either LoRA or RAG, both interventions are expected to influence it indirectly: RAG by supplying explicit retrieved passages that can

be cited in answers, and LoRA by training models on evidence-grounded gold answers. Attribution is therefore retained as a distinct dimension of groundness, evaluated alongside context relevance, faithfulness, and answer relevance. Together, these four dimensions provide a multidimensional perspective on answer quality and allow the framework to capture both the direct and indirect effects of different interventions.

3.2 Expected Relationships

Building on the constructs defined above, this study conceptualizes the relationship between interventions (RAG, LoRA, and their combination) and the outcome construct of groundness. Groundness, decomposed into four dimensions—context relevance, faithfulness, answer relevance, and attribution—serves as the lens through which the effects of these interventions are assessed. Prior research suggests that external augmentation and internal adaptation strategies influence these dimensions in complementary ways, and their combination may yield synergistic benefits.

Retrieval-Augmented Generation (RAG) is expected to primarily improve context relevance, as the retrieval of semantically related passages increases the likelihood that model outputs are anchored in appropriate evidence. By grounding responses in retrieved documents, RAG is also anticipated to support faithfulness, reducing reliance on unsupported parametric memory and mitigating hallucinations. Although attribution is not explicitly optimized, RAG provides explicit evidence that can be cited in answers, thereby indirectly encouraging more transparent outputs.

LoRA fine-tuning, by contrast, is expected to enhance faithfulness by aligning model outputs more closely with annotated gold-standard answers. Furthermore, by adapting parameters toward domain-specific supervision, LoRA is anticipated to improve answer relevance, reducing off-topic or under-informative content. Similar to RAG, LoRA may exert indirect effects on attribution, as training on evidence-grounded examples encourages models to generate responses that more consistently reflect their supporting sources.

Finally, when RAG and LoRA are combined, their effects are expected to be synergistic rather than merely additive. RAG supplies external knowledge that ensures access to verifiable and up-to-date evidence, while LoRA improves the model’s capacity to faithfully

integrate this evidence and generate relevant answers. Together, these interventions have the potential to enhance all four dimensions of groundness, making their combined application a more comprehensive strategy for addressing the challenges of long-form question answering.

3.3 Operationalization

To empirically assess the effects of the interventions on the four dimensions of groundness, this study relies on automated evaluation frameworks rather than human annotation. Each dimension is linked to established metrics that operationalize abstract constructs into measurable outcomes.

- **Context relevance** is measured using the RAGAS framework (“Faithfulness Metric in RAGAS,” 2023), which incorporates metrics such as context precision and recall. These metrics quantify the degree to which retrieved passages are semantically aligned with the query, allowing a systematic evaluation of whether RAG effectively improves evidence selection.
- **Faithfulness** is evaluated using RAGAS and the Attributable to Identified Sources (AIS) framework, which tests whether generated statements are attributable to supporting evidence (Rashkin et al., 2022). RAGAS assesses the proportion of generated claims that can be directly inferred from retrieved context, while AIS provides a complementary perspective by verifying whether statements in the output are attributable to identified sources. Together, these tools capture the extent to which model answers remain consistent with the supporting evidence.
- **Answer relevance** is assessed using semantic similarity measures. While AlignScore (Zha et al., 2023) was originally considered, version incompatibility led to its replacement with the e5-base-v2 embedding model (Cao & Wang, 2022). By computing embedding-based similarity between generated answers and gold-standard responses, this approach quantifies the degree to which outputs satisfy the user’s information need.
- **Attribution** is operationalized using the AIS framework, which explicitly tests whether model statements can be traced back to verifiable sources. This provides a direct measure of transparency, distinguishing answers that are genuinely grounded from those that rely on unsupported generation.

By aligning each dimension of groundness with specific evaluation metrics, the study establishes a systematic operationalization strategy. This ensures that interventions are not only compared at a holistic level but are also evaluated across distinct aspects of groundness, enabling a fine-grained analysis of where and how LoRA and RAG exert their effects.

In recent work, automated evaluation has been shown to approximate human judgment with high reliability. For example, [Zheng et al., \(2023\)](#) demonstrate that LLM-as-a-judge methods such as MT-Bench and Chatbot Arena achieve over 80% agreement with human evaluators, while [Li et al., \(2024\)](#) report correlations above 95% between LLM-based rankings and expert annotations in their BenchBuilder pipeline. These findings suggest that automated approaches, when carefully designed, can provide a scalable and cost-effective alternative to human evaluation without substantial loss in accuracy. Against this backdrop, the present study adopts automated metrics as the primary means of operationalizing groundness, balancing methodological rigor with feasibility.

3.4 Figure Framework and Summary

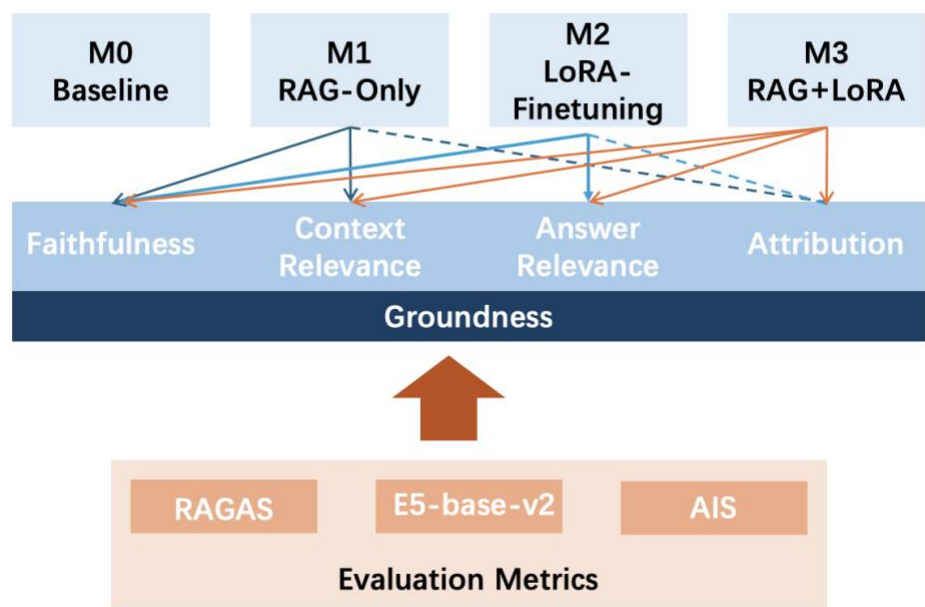


Figure 1 Conceptual Framework

Figure 1 presents the conceptual framework of this study. The interventions, LoRA fine-tuning and Retrieval-Augmented Generation (RAG), represent internal and external strategies for improving large language model performance. Each is expected to influence distinct dimensions of groundness: RAG primarily enhances context relevance and supports faithfulness, while LoRA strengthens faithfulness and answer relevance. Both interventions are also expected to indirectly affect attribution, and their combination may yield synergistic improvements across all four dimensions. Groundness is operationalized through automated evaluation metrics, including embedding-based similarity (e5, RAGAS), attribution measures (AIS), and semantic alignment scores (e5-base-v2).

4.Methodology

4.1 Research Onion

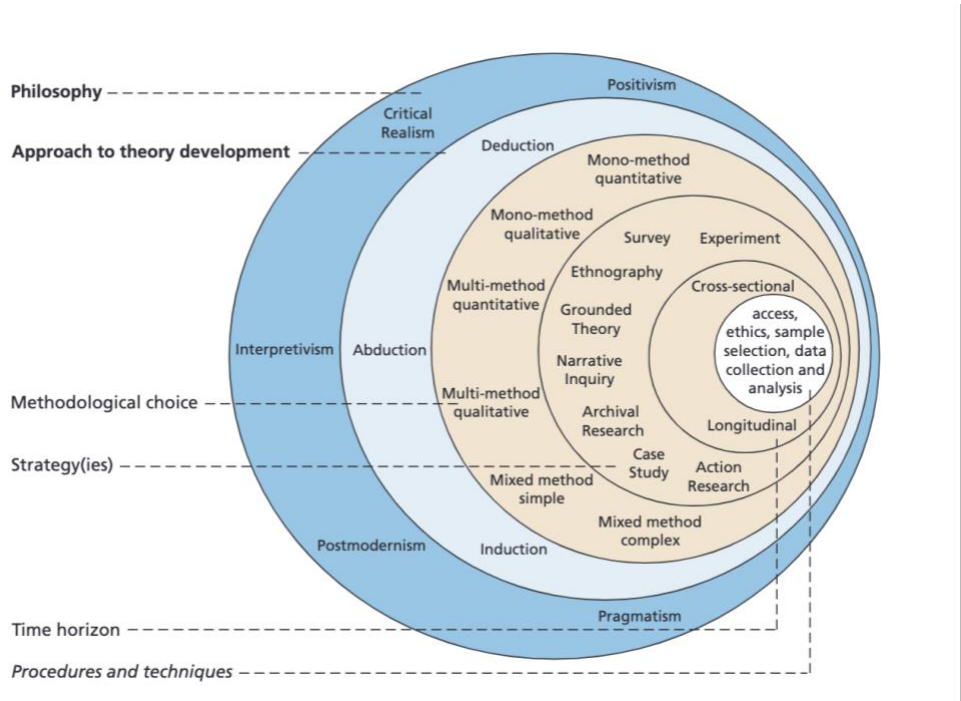


Figure 2 Research Onion

This thesis organizes its methodology with the “Research Onion” proposed by Saunders et al. (2019). The framework is applied to structure how knowledge is developed about improving answer groundness in evidence-based QA. Figure 2 visualizes the onion and its six layers—

research philosophy, research approach, methodological choice, research strategy, time horizon, and research techniques and procedures. We progress from the outer to the inner layers, increasing the level of detail and iterating when needed to ensure coherence. In brief, this study adopts a post-positivist and pragmatist research stance, applying a deductive approach to theory testing. The methodological choice is primarily quantitative, implemented through an experimental and comparative strategy in which four model configurations (M0–M3) are evaluated across two datasets (GovReport-QS and QASPER). The study follows a cross-sectional time horizon, and employs techniques including LoRA fine-tuning, retrieval-augmented generation, and automated evaluation using RAGAS, e5-base-v2 embeddings, and AIS. The following sections detail each layer.

4.1.1 Research Philosophy

This research follows [Saunders et al., \(2023\)](#) by viewing research philosophy as a set of assumptions and beliefs about how knowledge is developed. Making these assumptions explicit not only clarifies the study and its methods but also strengthens its credibility through internal consistency, leading to a coherent research design. This thesis adopts three core types of assumptions aligned with the Saunders framework:

Ontological Assumptions. The domain contains objective, document-grounded facts (e.g., statements in government reports and scientific articles). Model interventions (LoRA, RAG) are treated as real, manipulable factors that can cause observable differences in outputs.

Epistemological Assumptions. Knowledge is advanced by systematic measurement and hypothesis testing. Valid evidence consists of replicable, quantitative indicators, such as RAGAS (context/faithfulness/answer relevance), e5-base-v2 (Q+Context→Answer consistency), and AIS (attribution to identified sources).

Axiological Assumptions. The study values transparency, reproducibility, and the ethical use of sources. All prompts, splits, and configurations are disclosed; attributions to evidence are required; and claims are evaluated for traceability and integrity.

Positioned within Saunders' five philosophies for business and management research (positivism, critical realism, interpretivism, postmodernism, and pragmatism), this thesis

primarily adopts a positivist stance. Specifically, it is deductive, employs controlled comparative experiments (M0–M3), and relies on observable metrics with stated uncertainty. At the same time, a pragmatist overlay informs practical choices (e.g., dataset selection, toolchains) to efficiently answer the research question. However, the dominant logic remains positivist/deductive rather than interpretivist or postmodern. This philosophical stance provides the rationale for linking the research question (improving groundness) to the experimental strategy, quantitative methods, and evaluation procedures used throughout this thesis (Saunders et al., 2023).

It is important to note that this study's position leans more toward post-positivism than traditional positivism. While traditional positivism seeks to discover absolute, universal truths and assumes a researcher can observe the world with complete objectivity, post-positivism acknowledges that knowledge is a human construct, potentially biased, and that our understanding of reality is incomplete and fallible. Therefore, while this research uses rigorous quantitative methods and controlled experiments to approximate an objective truth, it also recognizes that these findings have inherent uncertainty (i.e., "metrics with stated uncertainty") and accepts that the researcher's own axiological assumptions can influence the research process.

4.1.2 Research Approach

According to [Saunders et al., \(2023\)](#), research can be developed using three primary approaches: deduction, induction, and abduction.

Deduction is a top-down process that relies on existing scholarly literature and established theories. Researchers derive testable hypotheses from theory and design a study to rigorously assess these hypotheses against empirical data, with the aim of corroborating or falsifying the initial propositions.

Induction is a bottom-up process typically used when the goal is to develop theory. It begins with systematic data collection about a phenomenon; through observation and analysis, patterns and themes are identified, forming the basis for a new theoretical framework.

Abduction combines elements of deduction and induction and involves an iterative movement between data and theory. Starting from puzzling or unexpected observations, researchers propose the best-fit explanations, refine or modify existing theory, and then test emerging explanations with supplementary data.

This thesis adopts a deductive approach. We begin with theory-driven hypotheses about how LoRA and RAG affect answer groundness across four dimensions—Context Relevance, Answer Faithfulness, Answer Relevance, and Attribution/Transparency (e.g., $M2 > M0$ and potential synergy in $M3$). These constructs are operationalized using RAGAS, AlignScore, and AIS. We then conduct controlled comparisons among four system variants (M0–M3) under a consistent dataset, prompting, preprocessing, and evaluation protocol, holding all factors constant except the presence of LoRA and/or RAG. Statistical inference (confidence intervals and effect sizes) is used to assess whether the data support the hypotheses. This structured, theory-testing methodology enables a rigorous and verifiable examination of the research question.

4.1.3 Methodological choice

The choice of a research methodology determines the type of data collected and the analytical approach (Saunders et al., 2023). Research methods are broadly categorized as quantitative (numerical data) or qualitative (non-numerical data).

Within these categories, a study can use a mono-method, which relies on a single data collection and analysis technique. A multi-method approach uses multiple techniques, but all within the same category (either quantitative or qualitative). A mixed methods approach, however, integrates both quantitative and qualitative techniques.

This thesis follows a mono-method quantitative design. Controlled experimental comparisons are conducted across four model variants (M0–M3), and answer groundness is quantified along four dimensions—Context relevance, Answer faithfulness, Answer relevance, and Attribution/Transparency—using RAGAS, AlignScore, and AIS. All analyses are performed

with a consistent dataset, prompting, preprocessing, and evaluation protocol, ensuring that only the presence of LoRA and/or RAG varies across conditions.

4.1.4 Research Strategy

A research strategy is a key element for successfully conducting research. It specifies the concrete actions required to answer a defined research question and serves as the methodological bridge between the underlying philosophy and the chosen methods (Saunders et al., 2023). In defining the strategy, the research question and goals are aligned with the study's philosophy, approach, and methods, while existing knowledge, data availability, and resource constraints also guide the choice. Saunders et al., (2023) discuss several strategies—experiment, survey, archival and documentary research, case study, ethnography, action research, grounded theory, and narrative inquiry.

An experimental strategy is primarily adopted for this thesis, as the objective is to obtain causal evidence about how architectural choices—LoRA and RAG—affect answer groundness across four dimensions (Context relevance, Answer faithfulness, Answer relevance, Attribution/Transparency) in long-document QA. Accordingly, controlled comparisons are run across four system variants (M0–M3), with all other factors held constant. This experimental design is supported by archival and documentary research; specifically, archival/documentary datasets (GovReport-QS as primary; QASPER as comparative) are used to provide reproducible, domain-relevant evidence.

4.1.5 Research Time Zone

A research time horizon is another key factor in shaping the design of a study. It determines whether a phenomenon is examined at a single point in time or over an extended period. As per Saunders et al., (2023), there are two primary time horizons: cross-sectional and longitudinal. A cross-sectional design provides a "snapshot" of a phenomenon at a specific moment, capturing data from a particular period without tracking changes over time. In contrast, a longitudinal design studies a phenomenon over a longer duration, enabling researchers to observe development and change.

This study employs a cross-sectional design. Concurrent, controlled comparisons are conducted among four model variants (M0–M3) using fixed archival datasets and a consistent evaluation protocol at a single period in time. No repeated time-based measurements are taken, and therefore, the design is not longitudinal.

4.1.6 Techniques and procedures

After having progressed through all the layers of the "Research Onion," the inner layer is reached. At this stage, more practical choices must be made, including the specific data collection techniques and analytical procedures required to meet the research objectives. As Saunders et al., (2023) note, it is critical to establish an alignment between this stage and all preceding ones to ensure a successful research setup.

In line with the chosen mono-method quantitative, experimental design, a clear and reproducible set of procedures was followed. The primary technique employed is a controlled experimental comparison of four distinct system variants—M0 (baseline), M1 (RAG only), M2 (LoRA fine-tuning only), and M3 (LoRA + RAG). To ensure the validity of these comparisons, all other variables, including the dataset, prompting strategy, preprocessing steps, and evaluation protocol, are held constant, thereby isolating the causal effects of the LoRA and RAG architectural choices.

For data and evaluation, archival and documentary sources are utilized, with fixed, publicly available datasets (primarily GovReport-QS, with QASPER for comparative analysis) serving as the basis for the study. The primary procedure for measuring the model outputs involves the application of a suite of quantitative, AI-assisted evaluation tools—RAGAS, e5-base-v2, and AIS—to quantify answer groundness across four specific dimensions.

Finally, the data analysis procedure involves applying statistical tests (e.g., confidence intervals and effect sizes) to the resulting quantitative scores. This rigorous process is used to test the initial theory-driven hypotheses and provide robust evidence to answer the research question.

4.2 CRISP-DM

The methodological design of this thesis follows a structured data science workflow, closely aligned with the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework (Wirth & Hipp, 2000). This ensures that the research process is systematic, transparent, and reproducible. The six CRISP-DM phases are reflected in the present study as follows.

- **Business Understanding:** The project goal is defined by the research questions, which investigate the individual and combined effects of LoRA fine-tuning and Retrieval-Augmented Generation (RAG) on the groundness of answers in long-document question answering tasks.
- **Data Understanding and Preparation:** We explored, cleaned, and deduplicated two benchmark datasets, QASPER and GovReport-QS, before unifying them into a standardized schema. A FAISS-based retrieval index was then constructed to support efficient retrieval.
- **Modelling:** Four distinct model variants were developed (M0 baseline, M1 RAG-only, M2 LoRA-only, and M3 hybrid RAG+LoRA). Each was designed to isolate and compare the contribution of different components.
- **Evaluation:** A multi-dimensional framework was used to assess the groundness of the answers. This was operationalized with established automated metrics like RAGAS, AlignScore, and AIS to measure context relevance, answer faithfulness, answer relevance, and attribution.
- **Deployment:** While this thesis does not focus on industrial deployment, the principles of the deployment phase are incorporated by ensuring the reproducibility of the research pipeline. This is achieved through the use of version-controlled scripts, standardized outputs, and methodological transparency, which facilitate the reuse of this work in future research and practice.

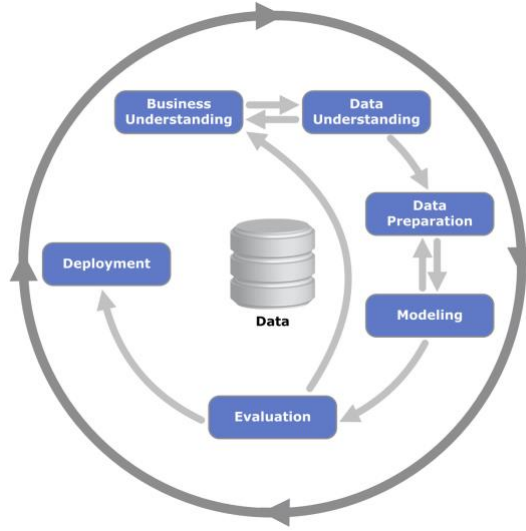


Figure 3 CRISP-DM

4.3 Data Understanding

This study employs two publicly available datasets designed for evaluating question answering (QA) systems in document-centric contexts: QASPER as a secondary dataset and GovReport-QS as the primary dataset. While both datasets involve long-form text and evidence-based answers, they differ substantially in domain, structure, and annotation protocols, making them complementary for evaluating large language models under different constraints.

4.3.1 GovReport-QS (Primary Dataset)

The GovReport corpus was first introduced for long-document summarization and contains over 19,000 reports published by the U.S. Congressional Research Service (CRS) and Government Accountability Office (GAO), many exceeding 10,000 words (Huang et al., 2021). To extend this corpus beyond summarization, GovReport-QS was proposed by (Cao & Wang, 2022) through the HIBRIDS framework. This extension reformulates the summarization task into a hierarchical question–summary (QS) benchmark, in which documents are paired with human-written questions, corresponding summaries, and annotated supporting spans.

The raw dataset expands into 25,055 question–answer pairs across 13 annotated fields. The official split follows a standard partition: 19,878 training pairs, 2,592 test pairs, and 2,585 validation pairs. The gold-standard answers are generally concise and fact-oriented, with the longest annotated answer reaching 1,106 characters.

Distinctive properties of GovReport-QS include:

- Domain: Formal government reports covering defence, healthcare, energy, economics, and other policy domains.
- Scale and retrieval challenge: Over 19,000 lengthy reports, with the QS subset including 6,153 annotated hierarchies, making retrieval particularly challenging since relevant evidence is sparse relative to document length.
- Annotation: Each QS pair includes both abstractive human answers and oracle evidence spans, suitable for evaluating retrieval-augmented QA as well as faithfulness metrics.

GovReport-QS is adopted as the primary dataset in this thesis, as it represents a realistic large-scale retrieval environment that strongly tests whether LoRA fine-tuning, and retrieval augmentation can improve groundness across context relevance, faithfulness, and attribution.

4.3.2 QASPER (Secondary Dataset)

QASPER (Dasigi et al., 2021) is a benchmark designed for document-grounded question answering over scientific research papers. It was created to reflect realistic information needs of researchers when reading academic publications. The dataset contains approximately 5,000 questions posed over 1,585 NLP papers drawn from Semantic Scholar.

The dataset is distributed as a DatasetDict with three splits: 888 training papers, 281 validation papers, and 416 test papers. Gold answers in QASPER are often long and discursive, reflecting the complex reasoning typical of scientific publications. The longest annotated gold answer reaches 2,223 characters, far longer than in GovReport-QS. Each answer is additionally linked to oracle evidence spans within the paper, making the dataset well-suited for evaluating faithfulness and attribution in large language model outputs.

Key features include:

- Question diversity: QASPER covers a wide spectrum of question types, ranging from factual (“Which dataset was used?”) to comparative or methodological (“How does the approach compare to prior work?”).
- Long-form answers: Human annotators provided abstractive gold answers, often multi-sentence or paragraph-level; the longest answers extend beyond 2,000 words, reflecting the complexity of academic discourse.
- Oracle evidence spans: Each answer is supported by manually identified passages from the source paper, enabling rigorous evaluation of both faithfulness and attribution.

QASPER is particularly useful for testing whether models not only generate plausible answers but also remain grounded in the source evidence. In this study, it is adopted as a comparative dataset, allowing assessment of whether methods generalize from government reports to the more technically dense scientific domain.

4.4 Data Preparation

Because GovReport-QS and QASPER differ in structure, two tailored preprocessing pipelines were applied, but both produced a consistent tabular format with question, gold answer, and oracle evidence fields, saved in CSV/Parquet for downstream use.

GovReport-QS. The JSONL files contain hierarchical question–summary structures. Processing steps included: (i) assigning files to train/validation/test splits, (ii) flattening nested questions into row-level entries with unique IDs, (iii) normalizing gold answers, (iv) extracting oracle evidence by traversing aligned section trees, (v) filtering rows without answers or evidence, and (vi) deduplication. The final dataset comprised 25,055 rows (19,878 train / 2,592 test / 2,585 validation), with a maximum gold answer length of 1,106 characters.

QASPER. Distributed via Hugging Face in Parquet format, QASPER required different cleaning rules. For each question, multiple annotations were normalized, with gold answers selected by majority preference (free-form > yes/no > extractive spans). Oracle evidence was drawn primarily from sentence-level highlights, otherwise from paragraph spans. Rows

without both an answer and evidence were removed, and duplicates dropped. The cleaned dataset yielded about 5,000 rows across 888 train / 281 validation / 416 test papers, with a maximum gold answer length of 2,223 characters.

In both datasets, preprocessing involved noise removal (filtering empty or duplicate entries), formatting and normalization (unifying answers and evidence text), and preserving the official train/validation/test splits. No explicit tokenization was applied at this stage, as that is handled by the downstream model tokenizer.

4.5 Modelling

4.5.1 Overview of Model Architecture

In the QASPER dataset, the gold-standard answers vary greatly in length, with the longest answer reaching 2,203 words. Due to computational constraints and the need for consistent evaluation across models, it is impractical to allow the language models to generate answers without a length limit. To address this, I apply a uniform generation cap of 192 tokens (approximately 120–150 words) to all models. This value strikes a balance between enabling concise, evidence-grounded answers and controlling inference time and cost. The prompts explicitly instruct models to respond concisely, ensuring that answers remain informative while avoiding abrupt truncation. By using a token-based limit rather than a word-based one, the restriction is consistent across languages and models, and it minimizes bias that could arise from differences in answer verbosity.

4.5.2 Model 0: Baseline

The baseline model (M0) employs the open-source *Mistral-7B-Instruct-v0.3* without any fine-tuning or retrieval augmentation. For each test instance, the model receives only the natural language question as input, without additional context. This setup establishes a lower-bound reference for evaluating subsequent improvements from LoRA fine-tuning and RAG.

To ensure consistency across experiments, the same decoding parameters are applied throughout: maximum generation length is capped at 192 tokens, greedy decoding is used to reduce variance, and random seeds are fixed for reproducibility. The model’s outputs are

stored alongside the corresponding question identifiers in the standardized schema, enabling direct comparison with other variants.

4.5.3 Model 1: RAG Only

The M1 model is a pure RAG pipeline designed to evaluate the performance of open-source components on question-answering tasks for scientific papers and government reports without any model fine-tuning. The pipeline operates in three main stages: (i) document preprocessing and indexing, (ii) retrieval with optional reranking, and (iii) generation. This unified design ensures a consistent experimental protocol across both datasets while respecting their unique structures.

Document Preprocessing and Indexing

To enable passage-level retrieval, all documents were segmented into paragraph-sized *chunks* of approximately 1,200 characters with an overlap of 200 characters. Each chunk was embedded using the `intfloat/e5-large-v2` model, applying the required query- and passage- prefixes and normalized vectors (`normalize_embeddings=True`). A FAISS `IndexFlatIP` was then constructed over these normalized embeddings, where the inner product approximates cosine similarity. To prevent data leakage, a separate index was built for each dataset split (e.g., QASPER-validation, QASPER-test, Government-validation), ensuring that the retrieval corpus for a given question was restricted to the text available in its corresponding split.

For the **QASPER dataset**, source documents included the title, abstract, and full text of scientific papers. Metadata such as `{paper_id, chunk_id}` was retained to ensure traceability. Tokens like `BIBREFxx`, which occasionally appear in gold-standard answers, were preserved to maintain alignment between retrieved evidence and reference annotations.

For the **Government Report dataset**, raw documents were sourced from the CRS and GAO JSON files. From each report, the title, summary or highlight, and the recursively flattened nested body were extracted. Only reports whose IDs were specified in the split identifiers (e.g., `split_ids/*valid*.ids`) were indexed. Metadata `{report_id, chunk_id}` was preserved. Although a fast JSON flattener and sentence-based chunker were applied, chunk size and overlap were kept consistent with the QASPER preprocessing, unless otherwise varied in ablation experiments.

Retrieval and Reranking Strategy

A same-document retrieval strategy was employed to ensure that retrieved context remained highly relevant and confined to the source document of the question. The process followed three steps: (i) a widened candidate pool (typically $K = 200\text{--}500$ passages) was retrieved from the FAISS index; (ii) candidates were filtered so that only passages originating from the same source document were retained; and (iii) the top- k passages ($k \in \{4, 5, 6\}$) were selected as the final context. For QASPER, filtering was performed on the basis of `paper_id`, while for Government Reports it was based on `report_id`.

An optional reranking step was introduced within the same-document pool to improve precision. A cross-encoder model (cross-encoder/ms-marco-MiniLM-L-6-v2) was used to rescore candidate passages, a step found particularly useful for list-based or “which-type” questions (e.g., datasets, methods, entities).

Generation and Output

In the final stage, the retrieved passages and the question were assembled into a structured prompt for the generator model. The prompt included the retrieved passages labeled as `[p1] ... [pk]`, together with a system instruction designed to enforce strict output rules: (i) answers must be derived exclusively from the provided context; (ii) extractive copying of spans or citation tokens (e.g., *BIBREF19*) was encouraged; (iii) “which/who/what” questions were answered with short, comma-separated lists; and (iv) when no sufficient information was available, the model was required to respond with “*I don’t know based on the provided context.*” An anchor token was appended to mark the boundary, and only newly generated tokens were decoded to prevent prompt echoing.

```
# 4) System instruction (extractive bias + short list answers + strict fallback)
SYS = (
    "You are a careful research assistant. "
    "Use ONLY the provided context to answer. "
    "Prefer EXTRACTIVE answers: copy exact spans, names, or citation tokens as they appear (e.g., BIBREF19). "
    "If the question asks 'which/who/what', output ONLY a short comma-separated list, no explanations. "
    "If the context is insufficient, say exactly: I don't know based on the provided context. "
    "Do NOT repeat the question, context, or any headers. "
    "Keep the answer under 192 tokens."
)
```

Figure 4 Prompt for M1

Answer generation was performed using the Mistral-7B-Instruct-v0.3 model under deterministic decoding (temperature = 0.0, top_p = 1.0), with a maximum length of 192 new tokens. Outputs were concise and could include bracketed passage citations (e.g., [p2]) to signal evidence use. For each question, the generated answer and the identifiers of the retrieved passages (*chunk_ids*) were stored. Gold-standard answers and oracle evidence were never used during indexing or prompting and served only for evaluation.

All M1 experiments followed a unified configuration: documents were chunked into ~1,200 characters with a 200-character overlap, embeddings were produced with *e5-large-v2* using normalized query:/passage: prefixed vectors, and FAISS IndexFlatIP indices were constructed separately for each dataset split. Retrieval expanded an initial candidate pool ($K = 200\text{--}500$), applied a same-document filter, and selected the top- k passages ($k \in \{4, 5, 6\}$). When reranking was enabled, the MiniLM-L-6-v2 cross-encoder rescored ~30 in-document candidates before final selection.

4.5.4 Model 2: Lora Fine-tuning Only

To isolate the effect of explicit evidence reading, a no-context control model (M2) was constructed. In this setting, the model receives only the question as input and is trained to generate the corresponding gold-standard answer. This configuration provides a controlled baseline against M0 and M1, while holding constant the backbone architecture, output length cap, and decoding parameters.

The training procedure applies LoRA-based supervised fine-tuning on the question–answer pairs from GovReport-QS and QASPER. Each question functions as an instruction, while the associated reference answer serves as the target output. This approach mirrors the instruction–response paradigm common in large-scale instruction tuning. However, unlike general-purpose models such as FLAN (Wei et al., 2022), the present setup restricts training to long-form QA tasks on specialized datasets. The objective is thus not broad instruction alignment but the targeted adaptation of the base model to improve faithfulness, answer relevance, and overall groundness in the QA domain.

Data & Preprocessing

The training data for M2 were drawn from the train splits of the QASPER and GovReport-QS datasets, while the validation splits were used for development and reporting. Only the *question* and *gold-standard answer* fields were retained. Answers underwent light normalization, including trimming of whitespace and flattening of simple lists or dictionary-like annotations into plain text. Importantly, no retrieval was applied: the model received only the question as input, with no passages or oracle evidence provided (“no-context” setting).

Each training instance was formatted into an instruction–response prompt (Figure 5). During optimization, supervised cross-entropy was applied with label masking: all tokens preceding the gold answer (i.e., up to and including the [/INST]\n marker) were masked with -100 and excluded from the loss. Only the answer span contributed to the training objective, ensuring that the model learned to produce the target output without overfitting to the prompt template and thereby stabilizing output style.

Each example is formatted as:

```
<s>[INST] <<SYS>>
You are a careful research assistant.
<</SYS>>
Question: {q}
Instruction: Answer concisely (≤120 words). If unsure, say you don't know.
[/INST]
{a}</s>
```

Figure 5 Prompt for M2

Backbone & Adapter

The M2 model was built on the Mistral-7B-Instruct-v0.3 backbone, run in bfloat16 precision. Parameter-efficient fine-tuning was implemented using the LoRA (PEFT) framework, with adapters inserted into both attention and MLP blocks (q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj). Unless otherwise specified, LoRA hyperparameters were set to $r = 16$, $\alpha = 32$, dropout = 0.05, and bias=none.

During training, use_cache was disabled to permit gradient flow, input gradients were enabled, and gradient checkpointing was optionally applied to manage memory constraints.

Training samples were rendered as full strings on CPU and pre-tokenized once into (input_ids, attention_mask). A masked SFT collator—functionally equivalent to TRL’s DataCollatorForCompletionOnlyLM—applied dynamic padding and constructed masked labels, ensuring that only the answer tokens contributed to the loss. To remain compatible with this masking scheme, packing was disabled.

Optimization & Hyperparameters

Training was performed with the AdamW optimizer, using the fused variant where available (preferred on A100 GPUs). The learning rate was set to 2×10^{-4} with a 3% warmup ratio followed by cosine decay. Models were trained for two epochs, with the global batch size determined by the product of micro-batch size and gradient accumulation steps to fit GPU memory constraints. The maximum sequence length was selected within the range of 768–1024 tokens, based on the observed distribution of question–answer pairs.

To ensure correct loss computation after pre-tokenization, remove_unused_columns=False was set in the Hugging Face Trainer. Optional performance optimizations included enabling torch.compile (with safe fallback to eager mode), Triton autotuning, and TensorFloat-32 (TF32) acceleration for eligible matrix multiplications.

Inference & Decoding

During inference, the tokenizer was configured to use left padding, following best practices for decoder-only architectures. To prevent prompt echoing, only tokens generated beyond the input length were decoded (i.e., generated[:, input_len:]). For comparability with M0 and M1, the maximum output length was capped at 192 tokens.

Two decoding presets were applied depending on the evaluation goal. For deterministic evaluation, greedy decoding was used (do_sample=False, num_beams=1). For exploratory purposes, an optional sampled variant was employed with stochastic decoding (do_sample=True, temperature = 0.2, top_p = 0.9, repetition_penalty ≈ 1.05), with no_repeat_ngram_size=3 optionally enforced to suppress verbatim repetition.

Engineering Considerations

Several engineering choices were made to ensure that M2 functioned as a stable and interpretable baseline. Comparability was maintained by using the same backbone architecture and a fixed 192-token output cap as in M0 and M1. Efficiency and reuse were supported by training only lightweight LoRA adapters, which can be saved, resumed, swapped, or merged into a single checkpoint for deployment. Robustness was promoted through masked supervised fine-tuning (to prevent prompt leakage), pre-tokenization (to avoid packing and collator conflicts), new-token decoding (to eliminate prompt artifacts), and left padding at inference (to improve generation efficiency). Portability was achieved by designing the pipeline to transfer seamlessly between QASPER and GovReport datasets, requiring only minimal changes in data paths or column names.

All experiments were conducted on a single A100-40GB GPU in *bfloat16* precision, with gradient checkpointing, fused AdamW, and optional torch.compile/Triton autotuning enabled as needed to balance memory and throughput. Overall, M2 serves as a low-cost, no-context supervised baseline for scientific and governmental QA, capturing instruction-following and learned parametric knowledge without external evidence. It thereby provides a clean control condition for evaluating future variants that incorporate retrieval or explicit evidence modeling.

4.5.5 Model 3:

The M3 model combines retrieval-augmented generation (RAG) with parameter-efficient fine-tuning (LoRA) in order to improve the groundness of model outputs across four dimensions: context relevance, faithfulness, answer relevance, and attribution. While M1 (pure RAG) relies solely on retrieval without adaptation, and M2 (pure LoRA) fine-tunes on oracle evidence that is not available at inference, M3 is designed to integrate both approaches. It enables the model to leverage retrieved passages in a realistic setting while benefiting from supervised fine-tuning.

Data Preparation

The training and evaluation pipeline is built on two datasets: GovReport-QS (primary) and QASPER (comparative). For each question, documents are segmented into passages of 1,500 characters with 200-character overlap. An embedding index is constructed using e5-large-v2 embeddings (Wang et al., 2022), and similarity search is performed with FAISS. To maintain document-level faithfulness, retrieval is restricted to passages from the same report ID. For each question, the top- k passages ($k = 8$) are selected as candidate contexts.

To ensure reproducibility, the retrieved passages are saved into a JSONL file, and this “frozen top- k ” is used for both validation and test inference. Oracle evidence from the original dataset is only used for evaluation and for comparison with M2.

Training Procedure

M3 builds upon the LoRA adapter obtained from M2. Specifically, the model inherits the parameter-efficient weights from M2, which were trained on oracle evidence, and continues fine-tuning using the retrieved top- k passages as context. The base model is Mistral-7B-Instruct-v0.3, augmented with LoRA adapters (Hu et al., 2021) applied to attention and feed-forward projection matrices (e.g., q_proj , v_proj , o_proj , up_proj). Training is performed as supervised fine-tuning (SFT).

```
<s>[INST] <<SYS>>
You are a careful research assistant. Use ONLY the provided sources.
Cite sources inline like [s1], [s2]. If evidence is missing, say you don't know.
<</SYS>>
[CONTEXT]
... retrieved passages ...
[/CONTEXT]

Question: {question}
Instruction: Answer concisely (≤120 words) and cite sources.
[/INST]
Answer: {gold_answer}
```

Figure 6 Prompt for M3

The model is trained to predict the gold answer while masking the prompt tokens. Optimization is performed with AdamW, learning rate $5e-5$, gradient accumulation, and BF16/4-bit quantization for efficiency.

Inference Procedure

During inference, questions are paired with their pre-retrieved passages from the frozen JSONL file. The passages are inserted into the same prompt format used during training. The model consists of the base Mistral-7B and the M3 LoRA adapter, which together generate answers limited to 192 tokens (≈ 120 words).

Generated answers are post-processed to remove template artifacts and are augmented with inline citations. Citations are attached to sentences by computing lexical overlap and, when enabled, semantic similarity with e5 embeddings. This ensures that each factual statement is traceable to one or more supporting passages.

4.6 Evaluation

To systematically assess the effectiveness of the four model variants (M0–M3), a multi-layered evaluation protocol was designed that ensures comparability across datasets and experimental settings. The evaluation setup focuses on the test split of both GovReport-QS (primary dataset) and QASPER (comparative dataset), with additional validation runs used only for model development and parameter tuning.

All reported results are based on the held-out test sets to avoid leakage from training or validation. Prior to evaluation, both datasets were cleaned to remove duplicated questions that would otherwise bias the retrieval and generation stages. After this process, the GovReport-QS test split contains 2592 unique question–answer pairs, while the QASPER test split includes 416 instances. Each question is paired with its gold-standard answer and oracle evidence passages, which serve as the reference for automatic metrics.

Final Schema

After preprocessing, the evaluation schema is standardized across datasets, with the following fields retained:

- **qid** — unique identifier of the question
- **doc_id** — source document identifier
- **question** — natural language question posed to the model
- **gold_answer** — human-annotated reference answer
- **oracle_context** — gold evidence passages used for evaluation
- **retrieved_context** — top-k passages returned by the retrieval module (for RAG variants)
- **model** — M0/M1/M2/M3
- **answer** — generated answers from each model variant.

Evaluation Dimensions and Metrics.

Groundness along four dimensions, each mapped to established automatic metrics:

- **Context relevance** — measured via retrieval-based metrics (RAGAS context precision/recall).
- **Answer faithfulness** — assessed using RAGAS faithfulness, which detects contradictions between generated answers and oracle evidence.
- **Answer relevance** — quantified with RAGAS answer relevancy and AlignScore. Due to version incompatibility with the original AlignScore implementation, the E5-base-v2 encoder was deployed as a substitute, which provides robust semantic alignment.
- **Attribution and transparency** — evaluated through the Answer Information Score (AIS), which measures whether supporting evidence is explicitly included or referenced in model outputs.

Reproducibility and Robustness

To reduce variance, random seeds are fixed at both retrieval and generation stages. All metrics are computed using open-source toolkits (e.g., RAGAS, AlignScore, AIS implementations), and code is maintained in a version-controlled environment to ensure replicability.

This evaluation protocol provides a consistent basis for comparing the baseline (M0), retrieval-only (M1), fine-tuned (M2), and hybrid (M3) models, and allows researchers to

directly test the research questions concerning the individual and joint contributions of LoRA fine-tuning and retrieval augmentation.

5. Results

5.1 Model Output

To illustrate the qualitative differences between the four model variants, one representative example from each dataset is presented.

5.1.1 GovReport-QS

Question: How have the Fed's crisis activities affected its net income?

Gold Answer: To date, the Fed's crisis activities have increased its net income.

Answers:

M0: crisis activities affected its net income? The Federal Reserve's crisis activities, such as large-scale asset purchases and lending programs, have increased its assets and liabilities, leading to an increase in its net income due to interest on these assets and fees from services provided. However, this increased income is often used to support economic stability rather than being distributed as profit.

M1: [p2], [p3] The provided context does not specify the effect of the Fed's crisis activities on its net income. However, it mentions that the Fed generates income primarily from the interest on their holdings of U.S. government securities, agency mortgage-backed securities, and agency debt acquired through open market operations. The Fed is required to transfer any surplus funds over \$7.5 billion to the General Fund of the U.S. Treasury. In 2016, the Federal Reserve transferred \$92 billion to the General Fund.

M2: The Federal Reserve System's (Fed's) net income for fiscal year 2010 was \$81.2 billion, down from \$95.9 billion in fiscal year 2009.

M3: To date, the Fed’s crisis activities have increased its net income. [s2] Two policy issues raised by the Fed’s actions are issues of systemic risk and moral hazard. [s2] Moral hazard refers to the phenomenon where actors take more risk because they are protected . [s2] The Fed’s involvement in stabilizing Bear Stear ... [s2]

Question	How have the Fed's crisis activities affected its net income?
Gold Answer	To date, the Fed's crisis activities have increased its net income.
M0 (Baseline)	Provides a long and partially accurate explanation, attributing income growth to assets and fees, but introduces additional speculation not grounded in the gold answer.
M1 (RAG)	States that the provided context does not specify the effect; instead reports unrelated details on the Fed’s income transfers, showing context anchoring but failing to answer the question.
M2 (LoRA)	Outputs specific figures for 2009–2010, which are factually correct but not responsive to the actual question, suggesting overfitting to training style.
M3 (RAG+LoRA)	Closely reproduces the gold answer (“increased its net income”) and cites source markers, though introduces extraneous discussion (moral hazard), partially diluting fidelity.

Table 1 Summary of a set of random generated answers from GovReport-QS

5.1.2 QASPER

Question: What languages are used as input?

Gold Answer: English

Answers:

M0:

used as input? The languages used as input can vary depending on the specific application or system. However, in general, most systems accept multiple languages such as English, Spanish, French, German, Chinese, Japanese, Arabic, and many more. It's essential to check the specific requirements for each system or application.

M1:

Hindi, Hausa, Turkish, French, Uzbek, Urdu

M2:

English, German, Spanish, Portuguese, French, Russian, Arabic, Chinese, Japanese, Korean, Dutch, Swedish, Italian, Polish, Turkish, Thai, Vietnamese, Finnish, Norwegian, Danish, Greek, Hebrew, Czech, Hungarian, Romanian, Slovak, Ukrainian, Indonesian, Malay, Afrikaans, Basque, Catalan, Cebuano, Icelandic, Kurdish, Latvian, Lithuanian, Serbian, Slovenian, Swahili, Tagalog, Urdu, Amharic, Somali, Tamil, Telugu, Burmese, Bengali, Hindi, Kazakh, Kyrgyz, Lao, Mongolian, Nepali, Sinhala, Tibetan, Yiddish, Zulu, Albanian, Bosnian, Croatian, Estonian

M3 :

English

Question	What languages are used as input?
Gold Answer	English
M0 (Baseline)	Produces a generic list of possible languages, typical of parametric memory without grounding.
M1 (RAG)	Outputs a short list of languages not supported by the gold answer, indicating overreliance on retrieved but irrelevant spans.

M2 (LoRA)	Produces an excessively long enumeration of languages, far beyond the gold answer, reflecting lack of precision in supervision.
M3 (RAG+LoRA)	Correctly outputs “English,” aligning exactly with the gold answer.

Table 2 Summary of a set of random generated answers from QASPER

5.2 Evaluation Metrics

Model performance was assessed along four dimensions of groundness: context relevance, faithfulness, answer relevance, and attribution. Each dimension was operationalized through specific automated metrics, as summarized in Table 3. Oracle-based scores use gold-standard evidence passages, while retrieval-based scores rely on passages returned by the retriever. In addition, embedding-based alignment (E5) was computed as a supplementary diagnostic to verify semantic consistency, but it is not part of the four primary dimensions.

Dimension	Metrics used	Notes
Context relevance	retrieval::context_precision, retrieval::context_recall	Only defined for models with retrieval (M1, M3).
Faithfulness	oracle::faithfulness, retrieval::faithfulness	Measures factual consistency of answers with evidence.
Answer relevance	oracle::answer_relevancy, retrieval::answer_relevancy	Captures whether the answer addresses the question.
Attribution	AIS (ais_oracle, ais_retrieval, aggregated ais)	Evaluates whether answers are attributable to identified sources.

Table 3 Dimensions and Evaluation Metrics

5.2.1 GovReport-QS

=== Oracle (mean) ===				
	oracle::answer_relevancy		oracle::faithfulness	
model				
M0	0.6073		0.3673	
M1	0.3450		0.3896	
M2	0.6130		0.5230	
M3	0.7458		0.6823	
=== Retrieval (mean) ===				
	retrieval::answer_relevancy		retrieval::faithfulness	retrieval::context_precision
				retrieval::context_recall
model				
M0	NaN	NaN	NaN	NaN
M1	0.3309	0.5547	0.3474	0.1210
M2	NaN	NaN	NaN	NaN
M3	0.5820	0.2723	0.6097	0.4333
=== E5 (mean) ===				
	e5_align_q_to_a	e5_align_q_oracle_to_a	e5_align_q_retrieval_to_a	
model				
M0	0.8830	0.8183	0.8830	
M1	0.8060	0.7865	0.8272	
M2	0.8553	0.8124	0.8553	
M3	0.8491	0.8385	0.8530	
=== AIS (mean) ===				
	ais_oracle	ais_retrieval	ais	
model				
M0	1.000	0.0000	0.0000	
M1	0.907	0.9066	0.9066	
M2	1.000	0.0000	0.0000	
M3	1.000	1.0000	1.0000	

Table 4 Evaluation metrics mean value for GovReport

Oracle-based metrics. Table 4 reports mean values across all evaluation metrics for the four model variants. Under the oracle-based metrics, M3 achieves the highest performance, substantially improving both answer relevance (0.746 vs. M0 = 0.607) and faithfulness (0.682 vs. M0 = 0.367). M2 also improves relative to M0, though with smaller gains in faithfulness (0.523). By contrast, M1 lags behind on oracle relevance (0.345) despite modest gains in faithfulness (0.390).

Retrieval-based metrics. For retrieval-based metrics, M3 outperforms M1 in answer relevance (0.582 vs. 0.331) and context precision/recall (0.610/0.433 vs. 0.347/0.121), indicating more effective utilization of retrieved evidence. However, M1 remains superior in retrieval-faithfulness (0.555 vs. 0.272), highlighting a trade-off between broader evidence coverage and strict consistency with retrieved content.

Embedding alignment (E5). The embedding-based alignment (E5) scores suggest relatively stable performance across models, with M0 and M2 slightly outperforming M1 and M3. This reflects that retrieval may introduce variability in semantic similarity to gold answers.

Attribution (AIS). Finally, the AIS attribution rate shows clear distinctions: M0 and M2 achieve full attribution under oracle evaluation but yield zero under retrieval (as no context was provided). In contrast, both M1 (0.907) and M3 (1.000) attain high attribution under retrieval, with M3 achieving perfect consistency.

Overall, these results show that M3 provides the strongest gains in relevance, recall, and attribution, but at the cost of reduced retrieval-faithfulness compared to M1. This pattern reinforces the central finding that combining retrieval with LoRA yields broader coverage and stronger grounding but introduces a coverage-consistency trade-off.

5.2.2 QASPER

=== Oracle (mean) ===				
	oracle::answer_relevancy	oracle::faithfulness		
model				
M0	0.6211	0.3497		
M1	0.5627	0.4677		
M2	0.6303	0.1744		
M3	0.7631	0.2504		
=== Retrieval (mean) ===				
	retrieval::answer_relevancy	retrieval::faithfulness	retrieval::context_precision	retrieval::context_recall
model				
M0	NaN	NaN	NaN	NaN
M1	0.5552	0.6489	0.8222	0.7949
M2	NaN	NaN	NaN	NaN
M3	0.7663	0.2200	0.5889	0.6677
=== E5 (mean) ===				
	e5_align_q_to_a	e5_align_q_oracle_to_a	e5_align_q_retrieval_to_a	
model				
M0	0.8853	0.7700	0.7700	
M1	0.8194	0.7405	0.7405	
M2	0.8130	0.7437	0.7437	
M3	0.7570	0.7094	0.7094	
=== AIS (mean) ===				
	ais_oracle	ais_retrieval	ais	
model				
M0	0.9975	NaN	0.9975	
M1	0.9441	0.9441	0.9441	
M2	1.0000	NaN	1.0000	
M3	0.6595	0.6595	0.6595	

Table 5 Evaluation Metrics Mean Value for QASPER

Oracle-based metrics. M3 achieves the highest answer relevance (0.763) but has reduced oracle faithfulness (0.250), indicating that while outputs are topically aligned with gold answers, consistency with gold evidence is weaker. M0 maintains stronger faithfulness (0.350) despite lower relevance. Both M1 (0.563 / 0.468) and M2 (0.630 / 0.174) show intermediate profiles, with M2 outperforming M1 in relevance but underperforming in faithfulness.

Retrieval-based metrics. Compared to M1, M3 substantially improves retrieval answer relevance (0.766 vs. 0.555) and recall (0.668 vs. 0.795), while sacrificing faithfulness (0.220 vs. 0.649). Context precision also decreases (0.589 vs. 0.822), confirming a trade-off: M3 uses broader evidence but adheres less strictly to retrieved content.

Embedding alignment (E5). All models perform relatively similarly in semantic similarity, though M0 retains the highest alignment with gold answers (0.885). M1 and M2 are slightly lower (0.819, 0.813), while M3 records the lowest values (0.757), suggesting that retrieval augmentation plus LoRA may reduce surface-level alignment when compared to direct generation.

Attribution (AIS). Clear distinctions emerge: M0 and M2 score near-perfect attribution under oracle evaluation (0.998–1.000) but yield no retrieval attribution since no evidence is provided. By contrast, M1 achieves high retrieval attribution (0.944), while M3 shows a reduction (0.660). This again reflects the coverage–consistency trade-off: M3 expands recall but weakens direct attribution to retrieved sources.

Overall, QASPER results contrast with GovReport findings. M3 delivers gains in answer relevance and recall but underperforms M1 in retrieval faithfulness and attribution. M0 remains competitive in both oracle faithfulness and embedding alignment, underscoring dataset-specific differences in how retrieval and fine-tuning interact.

5.2.3 Cross-dataset Observations

On GovReport-QS, M3 shows synergistic gains: LoRA boosts faithfulness, RAG provides attribution, and their combination yields the strongest oracle performance.

On QASPER, improvements are less consistent: RAG (M1) improves faithfulness and retrieval metrics, but LoRA (M2) degrades oracle faithfulness. The combined model (M3) optimizes answer relevancy but at the cost of faithfulness and attribution.

This highlights dataset-specific effects: GovReport-QS (policy reports) benefits more from LoRA fine-tuning, while QASPER (scientific papers) favours retrieval augmentation.

5.3 Statistical Analysis

The average scores reported in Section 5.2 provide an overview of model performance; however, such descriptive results do not establish whether the observed differences are statistically reliable.

Unit of analysis and pairing

The primary unit of analysis was the sample level (each question), with pairing defined by *paper_id* $\times$ *normalized question*. For robustness, results were also reported at the document level (each report): for each model, per-report scores were aggregated using the median (AIS was instead computed as the mean attribution rate), followed by paired testing. Owing to dataset structure, M2 vs. M0 comparisons were feasible primarily at the sample level, whereas M3 vs. M1 could be tested at both the sample and document levels. Comparisons involving M0 vs. M1/M3 and synergy tests were not performed due to insufficient overlap.

Tests and reporting

For continuous or bounded metrics (e.g., RAGAS scores, retrieval precision/recall, E5 alignment, document-level AIS rate), Wilcoxon signed-rank tests (Wilcoxon, 1945)&(Conover, 1999) were applied. Results are reported as Hodges–Lehmann median differences (Δ = later model - earlier model) with 95% bootstrap confidence intervals (Efron & Tibshirani, 1993) and effect sizes (Rosenthal, 1991). For binary AIS outcomes at the sample level, McNemar’s test (McNemar, 1947) was used, reporting odds ratios (OR) with 95% confidence intervals. At the document level, attribution was analysed via AIS attribution rates with Wilcoxon tests, avoiding degenerate McNemar cases caused by median aggregation

Multiple comparisons

Within each evaluation dimension (Answer relevance, Context relevance, Faithfulness, Attribution), all paired contrasts on the same dataset and analysis layer (sample vs. document) were adjusted using the Holm–Bonferroni procedure (Holm, 1979). Adjusted p-values (*p*_{adj}) are reported throughout.

Robustness checks

Sensitivity analyses included: (i) document-level aggregation using medians (or mean attribution rate for AIS), (ii) Winsorizing extreme values at the 1st/99th percentiles, and (iii) permutation testing by shuffling paired labels to re-estimate the distribution of Δ (Good, 2005).

5.3.1 GovReport-QS

Analyses were conducted at the report level to mitigate within-document dependence and variation in document length. For each metric, the per-report median was computed (with AIS evaluated as the mean attribution rate). Paired comparisons between M3 and M1 were then performed across reports ($n = 172$). For bounded continuous metrics, Wilcoxon signed-rank tests (two-sided) were applied, and Hodges–Lehmann median differences ($\Delta = M3 - M1$) with 95% bootstrap confidence intervals, effect sizes ($r = Z / \sqrt{N}$), and Holm–Bonferroni adjusted p-values were reported. For attribution, the report-level AIS rate was tested with Wilcoxon, while sample-level binary AIS was additionally analysed using McNemar’s test with odds ratios. Retrieval-dependent metrics were only defined for retrieval-enabled models, and pairs with missing values were excluded on a per-contrast basis. Report-level contrasts involving M0 and synergy analyses were not conducted on this dataset due to limited overlap.

Retrieval family (M3 vs M1, per-question)

At the sample level (per-question), M3 generally outperforms M1 in retrieval-related relevance and coverage: context recall shows a median improvement of +0.29 (95% CI: +0.25, +0.33), and retrieval-based answer relevancy improves by +0.03 (+0.01, +0.05). For context precision and AIS, the median difference is 0, but average gains (precision +0.26; AIS +0.09) indicate notable improvements on a subset of questions. In contrast, retrieval faithfulness decreases with a median difference of -0.30 (-0.33, -0.25), suggesting that M1 achieves higher consistency under strict retrieval-alignment scoring. This pattern is consistent with the document-level findings, highlighting a trade-off between improved relevance/coverage and reduced strict faithfulness to retrieved passages.

```

=== Quick summary (M3-M1 deltas; sample-level) ===

```

	metric	n	hl	ci_low	ci_high	mean_delta	median_delta
	ais	2589	0.0000	0.0000	0.0000	0.0935	0.0000
	ais_oracle	2589	0.0000	0.0000	0.0000	0.0931	0.0000
	ais_retrieval	2589	0.0000	0.0000	0.0000	0.0935	0.0000
	oracle::answer_relevancy	2589	0.0000	0.0000	0.5000	0.4001	0.0000
	oracle::faithfulness	2589	0.0000	0.0000	0.0000	0.2926	0.0000
	retrieval::answer_relevancy	2586	0.0309	0.0136	0.0454	0.2503	0.0309
	retrieval::context_precision	2585	0.0000	0.0000	0.0000	0.2634	0.0000
	retrieval::context_recall	2586	0.2857	0.2500	0.3333	0.3129	0.2857
	retrieval::faithfulness	2586	-0.3000	-0.3333	-0.2500	-0.2827	-0.3000

Table 6 Retrieved Family for GovReport

Retrieval family (M3 vs M1, per-report)

dimension	metric	n	test	hl	ci_low	ci_high	effect	effect_est	p_adj	dir
Answer relevance	oracle::answer_relevancy	172	Wilcoxon	-1.0000	-1.0000	-1.0000	r	-0.8767	0.0	M3↑
Answer relevance	retrieval::answer_relevancy	172	Wilcoxon	-0.8015	-0.8243	-0.4940	r	-0.7638	0.0	M3↑
Attribution	ais	172	McNemar	NaN	0.0198	50.4004	OR	1.0000	0.0	tie
Attribution	ais_oracle	172	McNemar	NaN	0.0198	50.4004	OR	1.0000	0.0	tie
Attribution	ais_retrieval	172	McNemar	NaN	0.0198	50.4004	OR	1.0000	0.0	tie
Context relevance	retrieval::context_precision	172	Wilcoxon	-0.5833	-0.6667	-0.4583	r	-0.7449	0.0	M3↑
Context relevance	retrieval::context_recall	172	Wilcoxon	-0.4000	-0.4643	-0.3500	r	-0.8290	0.0	M3↑
Faithfulness	oracle::faithfulness	172	Wilcoxon	-1.0000	-1.0000	-0.7000	r	-0.7658	0.0	M3↑
Faithfulness	retrieval::faithfulness	172	Wilcoxon	0.4724	0.3889	0.5345	r	0.7988	0.0	M1↑

Table 7 Paired tests (M1-M3) at the report level on GovReport-QS

Results show that M3 substantially improves answer relevance ($\Delta = +1.00$ and $+0.80$), context precision/recall ($\Delta = +0.58 / +0.40$), and oracle-based faithfulness ($\Delta = +1.00$), all with large effect sizes ($|r| \geq 0.74$, adjusted $p \approx 0$). In contrast, retrieved-evidence faithfulness favours M1 ($\Delta = -0.47$, $r = -0.80$), indicating a trade-off between more aggressive context use and strict consistency scoring. Report-level AIS attribution rates increase by $+0.083$ ($p \approx 9.3 \times 10^{-23}$). Sample-level paired results follow consistent directions and are provided in the appendix.

```

=== GovReport-QS: sample-level paired tests (M2 vs M0) ===

```

dataset	level	dimension	metric	contrast	n	test	hl	ci_low	ci_high	effect	effect_est	p_raw	dir	p_adj
GovReport-QS	sample	Answer relevance	oracle::answer_relevancy	M2-M0	2551	Wilcoxon	0.0	0.0	0.0	r	0.0	0.387	tie	0.387
GovReport-QS	sample	Faithfulness	oracle::faithfulness	M2-M0	2551	Wilcoxon	0.0	0.0	0.0	r	0.0	0.000	tie	0.000
GovReport-QS	sample	Attribution	ais	M2-M0	2551	Wilcoxon	0.0	0.0	0.0	r	NaN	NaN	tie	NaN
GovReport-QS	sample	Attribution	ais_oracle	M2-M0	2551	Wilcoxon	0.0	0.0	0.0	r	NaN	NaN	tie	NaN

Table 8 Paired tests (M0-M2) at the sample level on GovReport-QS

Paired Wilcoxon tests show no improvement in answer relevance ($\tilde{\Delta}=0$, 95% CI $[0,0]$, $r \approx 0$, $p=0.387$). For oracle-based faithfulness, the test is statistically significant ($p < 0.001$) but the

Hodges–Lehmann median difference is 0 with a [0,0] CI and $r \approx 0$, indicating no practically meaningful effect. AIS-based attribution exhibits no discordant pairs between M2 and M0 for most questions, yielding no informative McNemar contrast; descriptively, the models agree almost everywhere. Overall, M2 does not materially outperform M0 without retrieval on this dataset.

5.3.2 QASPER

While GovReport-QS emphasizes policy-oriented summaries with limited question–document alignment, the QASPER dataset consists of scientific articles with expert-annotated questions and gold evidence. Unlike GovReport-QS, where incomplete alignment led to insufficient overlap for some model contrasts (e.g., M0 vs. M1), QASPER does not suffer from this limitation. Each question is uniquely tied to a paper, ensuring that all models produce directly comparable outputs for the same set of questions. Consequently, all pairwise comparisons between models can be conducted at the question level, enabling a more comprehensive evaluation of attribution, answer relevance, context relevance, and faithfulness.

All Models, per-question

dimension	metric	contrast	test	n	p_raw	effect	effect_est	hl	ci_low	ci_high	a	b	c	d
Answer relevance	oracle::answer_relevancy	M1-M0	Wilcoxon	1213	0.000	r	-0.2162	-0.0198	-0.0336	-0.0055	NaN	NaN	NaN	NaN
Answer relevance	oracle::answer_relevancy	M2-M0	Wilcoxon	1213	0.000	r	-0.1801	-0.0446	-0.0537	-0.0345	NaN	NaN	NaN	NaN
Answer relevance	oracle::answer_relevancy	M2-M1	Wilcoxon	1212	0.025	r	0.0000	0.0000	0.0000	0.0000	NaN	NaN	NaN	NaN
Answer relevance	oracle::answer_relevancy	M3-M0	Wilcoxon	1214	0.171	r	-0.0393	-0.0612	-0.0747	-0.0471	NaN	NaN	NaN	NaN
Answer relevance	oracle::answer_relevancy	M3-M1	Wilcoxon	1213	0.000	r	0.2314	0.0061	0.0000	0.0201	NaN	NaN	NaN	NaN
Answer relevance	oracle::answer_relevancy	M3-M2	Wilcoxon	1213	0.002	r	-0.0896	-0.0027	-0.0110	0.0000	NaN	NaN	NaN	NaN
Answer relevance	retrieval::answer_relevancy	M3-M1	Wilcoxon	1210	0.000	r	0.2528	0.0118	0.0000	0.0248	NaN	NaN	NaN	NaN
Attribution	ais	M1-M0	McNemar	71	0.000	OR	19.5714	NaN	6.6857	57.2923	1145.0	68.0	3.0	0.0
Attribution	ais	M2-M0	McNemar	3	0.250	OR	0.1429	NaN	0.0074	2.7658	1213.0	0.0	3.0	0.0
Attribution	ais	M2-M1	McNemar	68	0.000	OR	0.0073	NaN	0.0005	0.1179	1148.0	0.0	68.0	0.0
Attribution	ais	M3-M0	McNemar	411	0.000	OR	823.0000	NaN	51.3885	13180.5596	802.0	411.0	0.0	3.0
Attribution	ais	M3-M1	McNemar	426	0.000	OR	9.5432	NaN	6.9041	13.1912	762.0	386.0	40.0	28.0
Attribution	ais	M3-M2	McNemar	414	0.000	OR	829.0000	NaN	51.7638	13276.4895	802.0	414.0	0.0	0.0
Attribution	ais_oracle	M1-M0	McNemar	71	0.000	OR	19.5714	NaN	6.6857	57.2923	1145.0	68.0	3.0	0.0
Attribution	ais_oracle	M2-M0	McNemar	3	0.250	OR	0.1429	NaN	0.0074	2.7658	1213.0	0.0	3.0	0.0
Attribution	ais_oracle	M2-M1	McNemar	68	0.000	OR	0.0073	NaN	0.0005	0.1179	1148.0	0.0	68.0	0.0
Attribution	ais_oracle	M3-M0	McNemar	411	0.000	OR	823.0000	NaN	51.3885	13180.5596	802.0	411.0	0.0	3.0
Attribution	ais_oracle	M3-M1	McNemar	426	0.000	OR	9.5432	NaN	6.9041	13.1912	762.0	386.0	40.0	28.0
Attribution	ais_oracle	M3-M2	McNemar	414	0.000	OR	829.0000	NaN	51.7638	13276.4895	802.0	414.0	0.0	0.0
Attribution	ais_retrieval	M3-M1	McNemar	426	0.000	OR	9.5432	NaN	6.9041	13.1912	762.0	386.0	40.0	28.0
Context relevance	retrieval::context_precision	M3-M1	McNemar	0	NaN	OR	NaN	NaN	NaN	NaN	0.0	0.0	0.0	143.0
Context relevance	retrieval::context_recall	M3-M1	Wilcoxon	1199	0.000	r	0.0000	0.0000	0.0000	0.0000	NaN	NaN	NaN	NaN
Faithfulness	oracle::faithfulness	M1-M0	Wilcoxon	1206	0.000	r	0.2315	0.0337	0.0000	0.1000	NaN	NaN	NaN	NaN
Faithfulness	oracle::faithfulness	M2-M0	Wilcoxon	1203	0.000	r	-0.3865	-0.1111	-0.1333	-0.0779	NaN	NaN	NaN	NaN
Faithfulness	oracle::faithfulness	M2-M1	Wilcoxon	1201	0.000	r	-0.5569	-0.2889	-0.3333	-0.2500	NaN	NaN	NaN	NaN
Faithfulness	oracle::faithfulness	M3-M0	Wilcoxon	1205	0.000	r	0.0000	0.0000	0.0000	0.0000	NaN	NaN	NaN	NaN
Faithfulness	oracle::faithfulness	M3-M1	Wilcoxon	1203	0.000	r	-0.3958	-0.1667	-0.2500	-0.1111	NaN	NaN	NaN	NaN
Faithfulness	oracle::faithfulness	M3-M2	Wilcoxon	1200	0.000	r	0.0000	0.0000	0.0000	0.0000	NaN	NaN	NaN	NaN
Faithfulness	retrieval::faithfulness	M3-M1	Wilcoxon	1198	0.000	r	-0.6322	-0.5000	-0.5000	-0.5000	NaN	NaN	NaN	NaN

Table 9 QASPER: SAMPLE-LEVEL paired tests (all model pairs)

Attribution (AIS). Retrieval substantially increased attribution scores. Compared to M0, M1 showed a highly significant gain (McNemar discordant pairs = 71; OR ≈ 19.6 , $p \approx 0$), and

M3 further improved over M1 (discordant = 426; OR $\approx$ 9.54, $p \approx 0$). The contrast M3 vs. M0 yielded an extremely large odds ratio (OR $\approx$ 823, $p \approx 0$). By contrast, LoRA alone (M2) performed worse: M1 clearly outperformed M2 (OR $\approx$ 0.007, $p \approx 0$), while M2 vs. M0 showed no significant difference ($p = 0.25$). These results confirm that retrieval and retrieval+LoRA strongly enhanced attribution, whereas LoRA-only lagged behind.

Answer relevance (oracle-based). In contrast to GovReport-QS, the baseline model M0 already performed strongly on oracle answer relevance. Both M1 (HL = -0.020, $r \approx -0.22$, $p \approx 0$) and M2 (HL = -0.045, $r \approx -0.18$, $p \approx 0$) scored significantly lower than M0. M3 improved modestly over M1 (HL = +0.006, $r \approx +0.23$, $p \approx 0$), but the difference relative to M0 was not significant ($p_{\text{adj}} = 0.171$). Thus, M3 recovered some relevance compared to M1, but did not surpass the strong M0 baseline.

Context relevance (retrieval-based). M3 provided small but consistent gains in retrieval-based answer relevancy (HL = +0.012, $r \approx +0.25$, $p \approx 0$). For context recall, most samples showed $\Delta = 0$, but the non-zero differences favoured M3, leading to overall significance ($p \approx 0$). For context precision, M1 and M3 produced identical outputs on all pairable samples (no discordant pairs), resulting in no observable difference. In summary, M3 improved retrieval relevance and recall over M1, whereas precision remained indistinguishable at the sample level.

Faithfulness. Oracle-based faithfulness showed mixed patterns: M1 improved slightly over M0 (HL = +0.034, $p \approx 0$), whereas M2 decreased relative to both M0 and M1 (HL = - 0.111 and - 0.289, $p \approx 0$). M3 performed worse than M1 (HL = - 0.167, $p \approx 0$) and was essentially on par with M0. Retrieval-based faithfulness, however, strongly favoured M1 over M3 (HL = - 0.5, $r \approx - 0.63$, $p \approx 0$). This indicates that retrieval improves faithfulness modestly, while LoRA undermines it, and M3 faces a trade-off: greater coverage and relevance at the cost of reduced consistency with retrieved passages, consistent with findings on GovReport-QS.

All Models, per-document

dataset	cluster	dimension	metric	contrast	test	n	p_raw	effect	effect_est	hl	ci_low	ci_high	p_adj	
QASPER	paper_id	Answer	relevance	oracle::answer_relevancy	M1-M0	Wilcoxon	404	0.000	r	-0.2456	-0.0540	-0.0716	-0.0342	0.000
QASPER	paper_id	Answer	relevance	oracle::answer_relevancy	M2-M0	Wilcoxon	404	0.014	r	-0.1218	-0.0545	-0.0660	-0.0385	0.029
QASPER	paper_id	Answer	relevance	oracle::answer_relevancy	M2-M1	Wilcoxon	405	0.004	r	0.1433	0.0083	0.0000	0.0214	0.014
QASPER	paper_id	Answer	relevance	oracle::answer_relevancy	M3-M0	Wilcoxon	404	0.003	r	-0.1458	-0.0729	-0.0874	-0.0486	0.014
QASPER	paper_id	Answer	relevance	oracle::answer_relevancy	M3-M1	Wilcoxon	405	0.000	r	0.3279	0.0182	0.0042	0.0375	0.000
QASPER	paper_id	Answer	relevance	oracle::answer_relevancy	M3-M2	Wilcoxon	405	0.074	r	-0.0888	-0.0065	-0.0132	0.0011	0.074
QASPER	paper_id	Attribution	AIS (doc-level rate)	M1-M0	Wilcoxon	405	0.000	r	0.0000	0.0000	0.0000	0.0000	0.000	
QASPER	paper_id	Attribution	AIS (doc-level rate)	M2-M0	Wilcoxon	405	0.102	r	0.0000	0.0000	0.0000	0.0000	0.102	
QASPER	paper_id	Attribution	AIS (doc-level rate)	M2-M1	Wilcoxon	405	0.000	r	0.0000	0.0000	0.0000	0.0000	0.000	
QASPER	paper_id	Attribution	AIS (doc-level rate)	M3-M0	Wilcoxon	405	0.000	r	-0.7091	-0.3333	-0.3333	-0.3333	0.000	
QASPER	paper_id	Attribution	AIS (doc-level rate)	M3-M1	Wilcoxon	405	0.000	r	-0.6286	-0.2500	-0.3333	-0.2500	0.000	
QASPER	paper_id	Attribution	AIS (doc-level rate)	M3-M2	Wilcoxon	405	0.000	r	-0.7118	-0.3333	-0.3333	-0.3333	0.000	
QASPER	paper_id	Faithfulness	oracle::faithfulness	M1-M0	Wilcoxon	404	0.000	r	0.3467	0.1571	0.0893	0.1889	0.000	
QASPER	paper_id	Faithfulness	oracle::faithfulness	M2-M0	Wilcoxon	405	0.000	r	-0.5335	-0.1625	-0.1875	-0.1181	0.000	
QASPER	paper_id	Faithfulness	oracle::faithfulness	M2-M1	Wilcoxon	404	0.000	r	-0.7022	-0.3333	-0.3889	-0.3000	0.000	
QASPER	paper_id	Faithfulness	oracle::faithfulness	M3-M0	Wilcoxon	404	0.000	r	-0.3180	-0.1000	-0.1402	-0.0625	0.000	
QASPER	paper_id	Faithfulness	oracle::faithfulness	M3-M1	Wilcoxon	403	0.000	r	-0.5523	-0.2708	-0.3333	-0.2500	0.000	
QASPER	paper_id	Faithfulness	oracle::faithfulness	M3-M2	Wilcoxon	404	0.002	r	0.0000	0.0000	0.0000	0.0000	0.002	

Table 10 QASPER: DOC-LEVEL paired tests (non-retrieval metrics + AIS rate)

At the document level, paired comparisons across 404–405 reports reveal distinct patterns across dimensions.

Answer relevance shows that M0 remains stronger than M1, M2, and M3 when measured against oracle evidence (e.g., M3 - M0 HL = -0.073, 95% CI [-0.087, -0.049]); however, M3 achieves a modest but significant improvement over M1 (HL = +0.018, 95% CI [0.004, 0.038]).

Faithfulness increases with retrieval, as M1 outperforms M0 (HL = +0.157, 95% CI [0.089, 0.189]), whereas LoRA alone reduces faithfulness relative to both M0 and M1. The combined model performs worse than M1 (HL = -0.271, 95% CI [-0.333, -0.250]) and also below M0 (HL = -0.100, 95% CI [-0.140, -0.063]).

For attribution, AIS rates favour M1 and penalize M3 relative to both M1 and M0 (e.g., M3 - M1 HL = -0.250; M3 - M0 HL = -0.333). It should be noted, however, that this document-level attribution analysis averages over model-specific question sets within each paper and may diverge from the per-question McNemar results. Taken together, these results confirm a coverage-consistency trade-off on QASPER: adding LoRA to retrieval yields relevance gains (M3 vs. M1) but comes at the cost of reduced document-level faithfulness and attribution, while M0 remains competitive in oracle-based relevance.

These divergences arise from differences in weighting (questions vs. documents), inconsistent pairing sets, and aggregation methods. Zero-inflation and small denominators at the

document level further amplify discrepancies. In addition, stronger coverage in M3 may reduce strict retrieval faithfulness, leading to trade-offs between relevance and consistency.

6. Discussion

6.1 Answer the Research Questions

RQ1. How does augmenting a model with retrieval (M1) affect context relevance and answer faithfulness compared to a base model without retrieval (M0)?

Retrieval augmentation consistently improves context relevance, as shown by higher context precision and recall in the mean results (e.g., GovReport precision +0.35, recall +0.12; QASPER precision 0.82, recall 0.79). Significance tests at both sample and document levels confirm that these differences are statistically reliable. For faithfulness, the effects are more modest: mean oracle-based faithfulness increases slightly on GovReport ($0.37 \rightarrow 0.39$) and significantly on QASPER (HL = +0.034, $p \approx 0$).

Thus, the evidence from both averages and Wilcoxon tests supports the conclusion that retrieval mainly strengthens context relevance and provides smaller, dataset-dependent gains in faithfulness.

RQ2. How does LoRA fine-tuning (M2) influence faithfulness and answer relevance compared to a base model (M0)?

LoRA’s effects differ sharply across datasets. On GovReport, mean scores show clear gains in oracle faithfulness ($0.37 \rightarrow 0.52$) and stable answer relevance, supported by significant Wilcoxon results. On QASPER, however, mean values and paired tests both indicate declines in answer relevance (HL = -0.045, $p \approx 0$) and faithfulness (HL = -0.111, $p \approx 0$).

Taken together, both average trends and significance tests demonstrate that LoRA improves grounding on GovReport but undermines it on QASPER, highlighting strong dataset dependence.

RQ3. How does combining retrieval augmentation and LoRA fine-tuning (M3) influence answer groundness—particularly faithfulness and attribution—compared to their individual contributions (M1 and M2)?

On GovReport, the combined model M3 achieves the highest mean scores in oracle faithfulness (0.68) and attribution (AIS = 1.0), with Wilcoxon tests confirming large, significant improvements over both M1 and M2 (e.g., $\tilde{\Delta} = +1.00$, $r \geq 0.74$, $p \approx 0$). On QASPER, McNemar tests show that M3 strongly improves attribution relative to both M0 and M1 (OR ≈ 9.5 –823, $p \approx 0$), while relevance improves modestly over M1 (HL = +0.006, $p \approx 0$). However, faithfulness declines relative to M1 (HL = -0.167, $p \approx 0$).

The combined evidence from means and significance tests supports a synthesis: M3 produces synergistic gains in GovReport but shows a coverage–consistency trade-off on QASPER, where broader attribution comes at the cost of reduced strict faithfulness.

6.2 Implications

6.2.1 Theoretical implications for LLM-based QA

The results provide a nuanced perspective on the theoretical underpinnings of grounding in Large Language Models (LLMs). First, our findings establish retrieval augmentation as a consistently robust mechanism for enhancing factual grounding, a central tenet of the Retrieval-Augmented Generation (RAG) paradigm. By consistently improving context relevance and, in some cases, faithfulness, retrieval proves its methodological value in mitigating the inherent limitations of models' static, parametric memory. This reinforces the theoretical importance of dynamic, external knowledge access for achieving reliable outputs.

Second, the study uncovers a critical theoretical trade-off in fine-tuning strategies. We find that parameter-efficient fine-tuning (PEFT), exemplified by LoRA, introduces a strong domain dependence for improving grounding. Its effectiveness is conditional on the dataset's structural properties, proving beneficial in structured, homogeneous domains (e.g., policy reports) but detrimental in heterogeneous scientific discourse. This divergence suggests that parameter adaptation is not a universal solution but rather a domain-specific hypothesis

requiring careful validation, challenging the notion of a one-size-fits-all approach to LLM grounding.

Finally, our work introduces a key theoretical insight into evaluation methodology itself. The divergence between sample-level and document-level analyses reveals that methodological choices in aggregation and weighting directly influence conclusions about model performance. This finding highlights a fundamental challenge in assessing model reliability, underscoring the necessity for a multi-level evaluation framework. By demonstrating that improvements can cluster within specific documents, our results argue that robust conclusions about grounding require testing across different aggregation scales to ensure validity.

Taken together, these findings refine our understanding of how to build reliable LLMs: not through a single, dominant strategy, but through a hybrid approach that combines the stability of retrieval with the domain-specific adaptability of fine-tuning, all while employing a rigorous, multi-faceted evaluation methodology.

6.2.2 Conceptual implications for groundness

The findings reinforce that groundness is a multidimensional construct rather than a unidimensional concept, encompassing a set of complementary but potentially competing criteria: context relevance, answer relevance, faithfulness, and attribution. Our results highlight the importance of this multi-faceted view, demonstrating that gains in one dimension may be offset by declines in another. This is exemplified by the coverage-consistency trade-off observed between M3 and M1, where superior contextual coverage does not guarantee proportional gains in faithfulness. This underscores the necessity of evaluating grounding as a holistic, multi-criteria problem, acknowledging the inherent tensions between its constituent components.

6.2.3 Practical implications for deployment

For real-world applications, the findings provide a clear directive for tailoring model design to specific domain needs. In structured, homogeneous corpora such as policy reports, a hybrid augmentation strategy combining retrieval with parameter-efficient fine-tuning (e.g., LoRA)

is a promising approach for decision-support systems, as it yields robust improvements in groundness. However, in heterogeneous domains like scientific discourse, retrieval alone may prove more reliable, given the observation that LoRA can reduce faithfulness. This divergence in performance underscores the critical importance of a domain-adaptive deployment strategy, suggesting that a one-size-fits-all approach to augmentation is insufficient. Instead, deployment strategies should carefully select and combine augmentation methods based on the unique characteristics of the dataset and domain to ensure optimal and reliable performance.

6.3 Limitations

This section highlights the limitations associated with this thesis, which have an effect on the findings.

6.3.1 Dataset coverage

The analysis relies on two datasets—GovReport-QS and QASPER—that differ in structure, domain, and annotation style. While this contrast provides useful cross-domain insights, the findings may not generalize to other domains, such as conversational QA, legal documents, or biomedical texts. A key limitation of using publicly available datasets is the potential for data contamination. Since these datasets might have been included in the pre-training corpora of large language models, the observed performance gains could, to an unknown extent, be influenced by the models' prior exposure to the test data. This phenomenon, often referred to as "memorization," complicates the assessment of true generalization capabilities. Additionally, incomplete document alignment in the GovReport dataset limited the feasibility of certain pairwise comparisons and constrained the scope of significance testing. This further restricts the generalizability of the conclusions.

6.3.2 Evaluation methodology

The evaluation rests on automated metrics such as RAGAS, embedding alignment, and AIS. Although these tools correlate with human judgments, they cannot fully capture nuanced aspects of grounding. Differences between sample-level and document-level aggregation also

demonstrate sensitivity to methodological choices. While robustness checks (e.g., bootstrap, alternative aggregation) were included, residual uncertainty remains about how well these measures reflect true user-perceived quality.

6.3.3 Model configuration and computational constraints

The study used a single base model (Mistral-7B) with parameter-efficient fine-tuning (LoRA) and a fixed retrieval setup. Computational limitations constrained model design choices: for example, the combined model (M3) used a frozen top-k retrieval configuration, which may have restricted retrieval flexibility and led to underestimation of potential performance. Similarly, lightweight design choices—such as limited fine-tuning epochs, truncated outputs, and reduced retrieval depth—were made to fit GPU and runtime budgets, but these trade-offs may have biased results in ways that do not reflect full-capacity model behaviours. In addition, an alternative LoRA configuration (M2-2 in very early-stage design), which would have been trained on oracle rather than retrieved context, was considered but not implemented due to resource constraints, exploring this design could have provided further insight into the role of context type in fine-tuning.

6.3.4 Scope of grounding dimensions

Groundness was operationalized through four dimensions—context relevance, answer relevance, faithfulness, and attribution. While this multidimensional framework captures core aspects of groundness, other facets such as calibration, uncertainty, and robustness to adversarial inputs were not evaluated.

6.4 Future Work

6.4.1 Broadening datasets and evaluation

Future work should extend beyond GovReport-QS and QASPER to cover additional domains such as conversational QA, legal documents, biomedical texts, and multi-lingual corpora. Such expansions would test whether the observed trade-offs generalize across diverse settings. To mitigate risks of data contamination and memorization, newly

constructed or held-out datasets should be prioritized, complemented by adversarial or synthetic test sets. Evaluation practices could also be strengthened by combining sample- and document-level perspectives, integrating human judgments to complement automated metrics, and adopting robustness checks (e.g., permutation tests, alternative aggregation schemes).

6.4.2 Exploring model architectures and training strategies

This study employed Mistral-7B with LoRA fine-tuning under computational constraints. Future work could investigate larger base models, alternative adaptation methods (adapters, instruction tuning, full fine-tuning), and end-to-end approaches where retriever and generator are trained jointly rather than as separate modules. Addressing current limitations—such as the frozen top-k retrieval design in M3—could provide a more realistic assessment of retrieval-augmented models. Other promising directions include dynamic context generation and selection, hybrid dense–sparse retrieval, and multimodal retrieval augmentation (integrating images, tables, or audio) to support richer QA scenarios.

6.4.3 Refining theoretical and conceptual understanding

Groundness should be further developed as a multidimensional construct. Future work could design mathematical models of trade-offs across enhancement strategies (retrieval, fine-tuning, reasoning), disentangle parametric knowledge from generative behavior to clarify the roots of hallucination, and improve explainability through new transparency metrics or visualization tools. This deeper theoretical exploration may yield a more principled understanding of how LLMs balance coverage, consistency, and attribution.

6.4.4 Extending applications and societal impact

Grounded QA should be evaluated in high-stakes domains such as law, healthcare, and finance, where accuracy, attribution, and faithfulness are critical. This requires domain-specific datasets and stricter evaluation protocols. Ethical considerations must also be addressed: retrieval may reduce hallucination but can amplify bias if source documents are skewed. Future research should therefore investigate mechanisms for bias detection in

retrieval pipelines and explore human–AI collaboration, where models defer to experts when uncertain and present multiple candidate contexts to support decision-making.

6.5 Recommendation

Building on the study’s core findings—including the consistent benefits of retrieval, the domain-specific effects of LoRA, and the trade-offs between coverage and faithfulness—this section provides specific recommendations for designing, evaluating, and applying similar research more effectively. These recommendations highlight actionable steps that could have been implemented to strengthen the present work, offering a clear roadmap for future investigations.

6.5.1 Research design

This study showed that retrieval consistently improves context relevance, while LoRA’s effects vary across domains. However, the analysis was limited to a single LoRA setup (M2), leaving open questions about alternative fine-tuning strategies. Future research should therefore include design variants that were considered but not implemented, such as training LoRA on oracle rather than retrieved context, to better isolate the role of context type in adaptation. Likewise, incomplete document–question alignment in GovReport restricted pairwise testing; more comprehensive alignment would strengthen the robustness of comparisons. Finally, since all experiments were based on Mistral-7B, diversifying model architectures and sizes would help establish whether the observed trade-offs generalize across models.

6.5.2 Evaluation

The results highlighted important trade-offs across grounding dimensions, showing that gains in relevance and recall can coincide with declines in faithfulness. Yet the evaluation relied exclusively on automated metrics, which—although efficient—cannot fully capture user-perceived quality. Human annotation should therefore be integrated in future work to validate automatic scores. Moreover, evaluation protocols should be designed from the outset to include both sample-level and document-level perspectives, avoiding aggregation artifacts

observed in this study. Additional dimensions such as calibration, robustness, and uncertainty estimation should also be incorporated to broaden the operational definition of groundness.

6.5.3 Practical considerations

The findings revealed that RAG combined with LoRA can yield strong gains in policy-related domains (GovReport) but unstable results in scientific QA (QASPER), indicating domain-specific implications for deployment. Future applied research should extend evaluations to high-stakes fields such as law and medicine, where stricter grounding and attribution standards are required. Responsible AI practices should also be incorporated, since retrieval may mitigate hallucinations but amplify biases embedded in source documents. Lastly, this study employed lightweight design choices (e.g., frozen top-k retrieval) to fit computational budgets; future work should more systematically analyze these trade-offs against more flexible alternatives to clarify the cost–benefit balance in real deployments.

7. Conclusion

This thesis set out to examine how retrieval augmentation and parameter-efficient fine-tuning contribute to improving the groundness of large language models in question answering. The central research questions focused on whether retrieval improves context relevance and faithfulness (RQ1), whether LoRA fine-tuning strengthens faithfulness and answer relevance relative to a base model (RQ2), and whether the combination of retrieval and LoRA produces synergistic improvements in groundness beyond their individual contributions (RQ3). To address these questions, four model configurations were implemented: a base model without augmentation (M0), a retrieval-augmented model (M1), a LoRA-fine-tuned model (M2), and a combined RAG+LoRA model (M3). The experiments were conducted on two datasets—GovReport-QS and QASPER—which differ substantially in domain, structure, and annotation, thereby enabling cross-domain comparison.

The study involved extensive data preparation to align, clean, and normalize both datasets, followed by systematic evaluation of model outputs across the four grounding dimensions: context relevance, answer relevance, faithfulness, and attribution. Answer generation was conducted using four model variants: the base model (M0) directly produced answers without

additional input; the retrieval-augmented model (M1) generated answers based on retrieved passages indexed from the datasets; the LoRA-fine-tuned model (M2) adapted the base model to the QA task using parameter-efficient fine-tuning; and the combined model (M3) integrated retrieval-augmented inputs with LoRA-adapted weights. To ensure comparability, outputs were truncated to a fixed token limit and generated under consistent decoding parameters. Automated evaluation metrics (RAGAS, embedding alignment, AIS) were then employed, and statistical significance was tested using Wilcoxon signed-rank tests and McNemar’s test. Analyses were conducted at both the per-question and per-document levels, enabling sensitivity checks against aggregation choices.

The results provide clear answers to the research questions. For RQ1, retrieval augmentation (M1) consistently improved context relevance across both datasets and modestly enhanced faithfulness, confirming the value of retrieval as a robust grounding strategy. For RQ2, LoRA fine-tuning (M2) showed domain-dependent effects: on GovReport, it improved oracle-based faithfulness and answer relevance, while on QASPER it reduced both dimensions relative to the base model. For RQ3, the combined model (M3) delivered synergistic gains on GovReport, significantly outperforming M1 and M2 in faithfulness, answer relevance, and attribution. On QASPER, however, the combined model achieved strong improvements in attribution and recall but at the cost of reduced faithfulness compared to M1, revealing a coverage–consistency trade-off.

The thesis makes several contributions. Theoretically, it advances the conceptualization of *groundness* as a multidimensional construct, demonstrating empirically that improvements in one dimension can entail reductions in another. Methodologically, it develops a multi-level evaluation framework that integrates automated metrics with statistical testing at both the sample and document level, while also employing large language models as judges to approximate human evaluation more efficiently and consistently. Practically, it provides guidance for QA system deployment: combining retrieval with LoRA is promising for structured policy-related corpora, whereas retrieval alone may be more reliable in heterogeneous scientific QA.

The study is limited by its reliance on two datasets, the potential risk of data contamination, and incomplete document alignment in GovReport. Model design choices were constrained

by computational resources, including reliance on a single base model (Mistral-7B), lightweight LoRA configurations, and frozen top-k retrieval in M3. Evaluation was restricted to automated metrics without human annotation. Future work should broaden datasets to high-stakes domains, adopt more flexible retrieval architectures, incorporate human evaluation, and refine the theoretical modelling of groundness, including calibration and robustness.

In conclusion, this thesis demonstrates that retrieval is a consistently effective tool for grounding LLMs, while LoRA’s impact is domain-dependent, and their combination can produce either synergy or trade-offs depending on context. These findings reinforce that groundness is not a single property but a balance of dimensions—relevance, faithfulness, and attribution—that may compete with one another. The study contributes both empirical evidence and methodological insights, while also pointing toward future research needed to strengthen the design and deployment of grounded question answering systems.

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Appendix

Code and Data Outputs

The full source code used in this thesis, together with the generated answer sets, are provided in compressed files.

Code.zip contains all scripts used for preprocessing, fine-tuning, retrieval, and evaluation.

Outputs.zip contains the generated answers from all four model configurations (M0–M3) across both datasets.

These files (combined as appendix.zip) are submitted alongside the thesis and can be accessed in the electronic submission package.